MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, REGULATING, PULSE-WIDTH MODULATOR, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, pulse-width modulator microcircuits. Two product assurance classes and a choice of lead finishes are provided for each type and are reflected in the complete part number.

1.2 Classification.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Regulating, pulse-width modulator
02	Regulating, pulse-width modulator
03	Regulating, pulse-width modulator
04	Regulating, pulse-width modulator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, .840" x .310" x .200"), dual-in-line package
V	D-6 (18-lead, .960" x .310" x .200"), dual-in-line package

1.3 Absolute maximum ratings.

Input voltage	42 V
Collector supply voltage:	
Device type 01	42 V
Device types 02, 03, and 04	37 V
Output current (each output):	
Device type 01	±100 mA
Device types 02, 03, and 04	±200 mA
Reference output current	-50 mA
Oscillator charging current	-5 mA
Logic inputs	-0.3 V to V_{REF}
Analog inputs	-0.3 V to V_{IN}
Lead temperature (soldering, 10 seconds)	+300°C
Storage temperature range	-65°C to +150°C
Junction temperature (T_J)	+150°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.4 Recommended operating conditions.

Input voltage range:	
Device type 01	+8 V to +40 V, class B +8 V to +40 V, class S
Device types 02 and 04 1/	+8 V to +30 V, class B +8 V to +27 V, class S
Device type 03 1/	+8 V to +20 V, class B +8 V to +20 V, class S
Oscillator frequency range:	
Device type 01	10 kHz to 200 kHz
Device types 02 and 04	150 Hz to 300 kHz
Device type 03	100 Hz to 350 kHz
Collector supply voltage:	
Device type 01	+4.5 V to +40 V
Device types 02, 03, and 04	+4.5 V to +35 V
Operating temperature range	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
16-lead dual-in-line	E	250 mW at T_A = +125°C	60°C/W	100°C/W
18-lead dual-in-line	V	250 mW at T_A = +125°C	42°C/W	100°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

1/ +8 V to +35 V operation is allowed if the maximum junction temperature is not exceeded. Heat sinking may be required.

TABLE I. Electrical performance characteristics.
(device type 01)

Characteristic	Symbol	Conditions V_{IN} = 20 V dc, R_T = 2 kΩ, ±0.1% C_T = 0.01 µF, ±0.1% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Limits Min	Max	Unit
Error amplifier					
Input offset voltage	V_{IO}	V_{CM} = 2.5 V	-5	5	mV
Input bias current	I_{IB}	V_{CM} = 2.5 V	.01	10	µA
Input offset current	I_{IO}	V_{CM} = 2.5 V	-1	1	µA
Compensation current (sink)	I_{COS1}	$V_{IN}(I)-V_{IN}(NI) \geq 150$ mV T_C = +25°C	65	170	µA
Compensation current (source)	I_{COS2}	$V_{IN}(NI)-V_{IN}(I) \geq 150$ mV T_C = +25°C	-170	-65	µA
Common mode rejection	C_{MR}	1.8 V ≤ V_{CM} ≤ 3.4 V	70		dB
Open loop voltage gain	A_{VS}	T_C = +25°C	72		dB
		T_C = -55°C, +125°C	68		dB
Unity gain bandwidth	G_{BW}	T_C = +25°C, see figure 7	2		MHz
Reference					
Output voltage	V_{REF}		4.8	5.2	V
Line regulation	V_{RLINE}	8 V ≤ V_{IN} ≤ 40 V	-20	20	mV
Load regulation	V_{RLOAD}	-20 mA ≤ I_{REF} ≤ 0 mA	-50	50	mV
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$	V_{IN} = 20 V, e_i = 1 V rms, f = 2400 Hz, T_C = +25°C	50		dB
Short circuit current limit	I_{OS} 1/	t ≤ 25 ms, T_C = +125°C	-120		mA
		V_{REF} = 0 V, T_C = -55°C, +25°C	-190		mA
Output noise voltage	N_O	T_C = +25°C 10 Hz ≤ f ≤ 10 kHz		200	µV rms
Oscillator					
Maximum frequency	f_{OSC} (MAX)	RT = 2 kΩ, ±0.1%	250		kHz
		C_T = 0.001 µF, ±1.0%			
Initial frequency	f_{OSC1}	T_C = +25°C	47	58	kHz
Oscillator	f_{OSC2}	T_C = +125°C	45	60	kHz
Frequency	f_{OSC3}	T_C = -55°C	45	60	kHz
Frequency change with voltage	$\frac{\Delta f_{OSC1}}{\Delta V_{IN}}$	8 V ≤ V_{IN} ≤ 40 V, T_C = +25°C	-2	2	%
Output amplitude	V_{OSC}		2.4		V
Output pulse width	t_{pw}		0.3		µs
Ramp voltage	V_{RAMP}		3.0	3.8	V
Saturation voltage	V_{SAT} (OSC)	I_{CT} = 5 mA, V_{OSC} = 5 V, T_C = +25°C	0.7	1.1	V

See footnotes at end of table.

TABLE I. Electrical performance characteristics - Continued.
(device type 01)

Characteristic	Symbol	Conditions V_{IN} = 20 V dc, R_T = 2 kΩ, ±0.1% C_T = 0.01 µF, ±0.1% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Limits Min	Limits Max	Unit
Comparator					
Duty cycle adjust range	$\frac{t_{ON}}{t_{OSC}}$ 2/	Min value: V_{CM} = 2.5 V, V_{comp} = 0.5 V		0.001	%
		Max value: V_{CM} = 2.5 V, V_{comp} = 3.8 V	45	50	%
Output section 3/					
Collector leakage current	I_{CEX}	V_C = 40 V		10	µA
Saturation voltage	V_{SAT}	$V_{IN}(NI) - V_{IN}(I) \geq 150$ mV, I_C = 50 mA		2.0	V
Emitter output voltage	V_{EO}	$V_{IN}(NI) - V_{IN}(I) \geq 150$ mV, T_C = +25°C, V_C = 20 V, I_E = -50 mA	17		V
Rise time	$t_{R(tr)}$	$V_{IN}(NI) - V_{IN}(I) \geq 150$ mV, R_C = 2 kΩ, V_C = 20 V, T_C = +25°C, C_L = 15 pF		0.4	µs
Fall time	$t_{R(tf)}$	$V_{IN}(NI) - V_{IN}(I) \geq 150$ mV, R_C = 2 kΩ, V_C = 20 V, T_C = +25°C, C_L = 15 pF		0.2	µs
Shutdown circuit					
Sense voltage	V_{SEN}	$V_{IN}(NI) - V_{IN}(I) \geq 150$ mV, V_{EOA} = 2 V, T_C = +25°C	190	210	mV
		T_C = +125°C, -55°C	165	235	mV
Shutdown 4/	V_{SD} (HI)	$V_{IN}(NI) - V_{IN}(I) \geq 150$ mV	1.4		V
	V_{SD} (LO)	$V_{IN}(NI) - V_{IN}(I) \geq 150$ mV		0.4	V
Total standby current					
	I_{IN}	V_{IN} = 40 V		10	mA

See footnotes at end of table.

TABLE I. Electrical performance characteristics - Continued.
(device types 02 and 04)

Characteristic	Symbol	Conditions V_{IN} = 20 V dc, R_T = 3.6 kΩ, ±0.1% R_D = 0 kΩ, C_T = 0.01 µF, ±0.1% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Limits Min	Limits Max	Unit
Reference section					
Output voltage	V_{REF}	T_C = -55°C, +125°C	5.0	5.2	V
		T_C = +25°C	5.05	5.15	V
Line regulation	V_{RLINE}	V_{IN} = 8 V to 35 V	-30	30	mV
Load regulation	V_{RLOAD}	I_{REF} = 0 to -20 mA	-50	50	mV
Short circuit current	I_{OS} 1/	V_{REF} = 0 V, t < 25 ms	-125		mA
Output noise voltage	N_O	10 Hz < f < 10 kHz		200	µV rms
Ripple rejection	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$	V_{IN} = 20 V, +1 V rms, f = 2400 Hz T_C = +25°C	50		dB
Oscillator section					
Initial frequency	f_{OSC}	T_C = +25°C	37.5	42.5	kHz
Oscillator frequency over temperature	f_{OSC2}	T_C = -55°C, +125°C	36	44	kHz
Saturation voltage	V_{SAT} (OSC)	I_{DIS} = 5 mA, V_{OSC} = 5 V, T_C = +25°C	0.5	1.1	V
Clock amplitude	V_{OSC}		3.0		V
Clock pulse width	t_{pw}	T_C = -55°C, +25°C	0.3	1.0	µs
		T_C = +125°C	0.3	1.4	µs
Ramp voltage	V_{RAMP}		3.0	3.6	V
Voltage stability	$\frac{\Delta f_{OSC1}}{\Delta V_{IN}}$	V_{IN} = 8 V to 35 V, T_C = +25°C	-1	1	%
Minimum frequency	f_{OSC} (MIN)	R_T = 150 kΩ, ±0.1% C_T = 0.1 µF, ±1.0%, R_D = 0 kΩ		150	Hz
Maximum frequency	f_{OSC} (MAX)	R_T = 2 kΩ, ±0.1% C_T = 0.001 µF, ±1.0%, R_D = 0 kΩ	300		kHz
Sync 4/	V_{SYNC} (LO)			0.8	V
	V_{SYNC} (HI)		2.8		V
Sync input current	I_{SYNC}	Sync voltage = 3.5 V		2.5	mA
Error amplifier section					
Input offset voltage	V_{IO}		-5	5	mV
Input bias current	I_{IB}		.01	10	µA
Input offset current	I_{IO}		-1	1	µA

See footnotes at end of table.

TABLE I. Electrical performance characteristics - Continued.
(device types 02 and 04)

Characteristic	Symbol	Conditions V_{IN} = 20 V dc, R_T = 3.6 kΩ, ±0.1% R_D = 0 kΩ, C_T = 0.01 µF, ±0.1% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Limits		Unit
			Min	Max	
Open loop voltage gain	A_{VS}	V_{CM} = 2.5 V	60		dB
Common mode rejection	C_{MR}	V_{CM} = 1.5 V to 5.2 V	60		dB
Supply voltage rejection ratio	P_{SRR}	V_{IN} = 8 V to 35 V	60		dB
Unity gain bandwidth	G_{BW}	A_V = 0 dB, see figure 7, T_C = +25°C	2		MHz
Output high level	V_{HI}		3.8		V
Output low level	V_{LO}			0.5	V
P.W.M. comparator section					
Maximum duty cycle	t_{ON}^{MAX}/t_{OSC}	2/ V_{COMP} = 3.6 V	45	50	%
Minimum duty cycle	t_{ON}^{MIN}/t_{OSC}	2/ V_{COMP} = 0.6 V		0.001	%
Output section 5/					
Output low	V_{OL}	I_{SINK} = 20 mA		0.4	V
		I_{SINK} = 100 mA		2.0	V
Output high	V_{OH}	I_{SOURCE} = -20 mA	18		V
		I_{SOURCE} = -100 mA	17		V
Under voltage lockout	V_{UL}		6	8	V
Shutdown delay	t_{SD}	V_{SD} = 3 V, T_C = -55°C, +25°C		500	ns
		T_C = +125°C		700	ns
Rise time	$t_{R(tr)}$			600	ns
Fall time	$t_{R(tf)}$			300	ns
V_C standby current	I_C	V_C = 35 V (device type 02 only)		200	µA
Soft start section					
Soft start current	I_{SS}	V_{SD} = 0 V	-80	-25	µA
Shutdown input current	I_{SD}	V_{SD} = 2.5 V		1	mA
Soft start voltage	V_{SS}	V_{SD} = 2 V		0.6	V
Shutdown voltage 4/	$V_{SD(LO)}$			0.5	V
	$V_{SD(HI)}$		1.6		V
Total supply current	I_{IN}	V_{IN} = 35 V, T_C = -55°C, +25°C		20	mA
		T_C = +125°C		18	mA

See footnotes at end of table.

TABLE I. Electrical performance characteristics.
(device type 03)

Characteristic	Symbol	Conditions V_{IN} = 15 V dc, R_T = 4.12 kΩ, ±0.01% R_D = 0 kΩ, C_T = 0.01 µF, ±0.1% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Limits Min	Max	Unit
Reference section					
Reference output voltage	V_{REF}		4.9	5.1	V
Line regulation	V_{RLINE}	V_{IN} = 8 V to 35 V	-20	20	mV
Load regulation	V_{RLOAD}	I_{LOAD} = 0 mA to ±20 mA	-30	30	mV
Short circuit current	I_{OS} 1/	V_{REF} = 0 V, t < 25 ms	-125		mA
Output noise voltage	N_O	10 Hz ≤ f ≤ 10 kHz, T_C = +25°C		200	µV rms
Ripple rejection	ΔV_{IN}	V_{IN} = 15 V, +1 V rms, T_C = +25°C Sinewave at 2.4 kHz	50		dB

TABLE I. Electrical performance characteristics - Continued.
(device type 03)

Characteristic	Symbol	Conditions V_{IN} = 15 V dc, R_T = 4.12 kΩ, ±0.01% R_D = 0 kΩ, C_T = 0.01 µF, ±0.1% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Limits Min	Limits Max	Unit
P.W.M. comparator					
Maximum duty cycle	$t_{ON}{}^{MAX}$	2/ V_{COMP} = 3.6 V	45	50	%
Minimum duty cycle	t_{OSC} $t_{ON}{}^{MIN}$ t_{OSC}	2/ V_{COMP} = 0.4 V		0.001	%
Output drivers 7/					
Saturation voltage	$V_{CE(SAT)}$	V_C = 15 V, I_{SINK} = 20 mA		0.3	V
		I_{SINK} = 100 mA		2.0	V
Output high	V_{OH}	V_C = 15 V, I_{SOURCE} = -20 mA	12.5		V
		I_{SOURCE} = -100 mA	12.0		V
Shutdown delay	t_{SD}	V_C = 15 V, T_C = -55°C, +25°C		0.5	µs
		T_C = +125°C		0.7	µs
Rise time	$t_{R(tr)}$	V_C = 15 V		0.3	µs
Fall time	$t_{F(tf)}$	V_C = 15 V		0.2	µs
V_C standby current	I_C	V_C = 35 V		150	µA
Digital ports: $\overline{SYNC}$, $\overline{RESET}$, $\overline{SHUTDOWN}$					
High input current	I_{IH}	V_{IH} = 2.4 V	-200		µA
Low input current	I_{IL}	V_{IL} = 0.4 V	-360		µA
High output voltage	V_{OHP}	I_{SOURCE} = 40 µA	2.4		V
Low output voltage	V_{OLP}	I_{SINK} = 3.6 mA		0.4	V
Current limit comparator					
Sense voltage	V_S		80	120	mV
Input bias current	I_{IBS}		-10		µA
Soft start section					
Error clamp voltage	V_{EC}			0.4	V
Capacitor charging current	I_{CS}		-150	-50	µA

See footnotes at end of table.

TABLE I. Electrical performance characteristics - Continued.
(device type 03)

Characteristic	Symbol	Conditions V_{IN} = 15 V dc, R_T = 4.12 kΩ, ±0.01% R_D = 0 kΩ, C_T = 0.01 µF, ±0.1% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Limits Min	Limits Max	Unit
Under voltage lockout					
Reset output (low) voltage	V_R (LO)	V_{REF} = 3.8 V		0.4	V
Reset output (high) voltage	V_R (HI)	V_{REF} = 4.8 V	2.4		V
Power consumption					
Standby current	I_{IN}	$\overline{SHUTDOWN}$ = 0.4 V, V_{IN} = 35 V T_C = -55°C, +25°C T_C = +125°C		30 25	mA mA

1/ Continuous short circuit limits will be less than indicated test limits.

2/ t_{OSC} is the period of the output waveform.

3/ Each output transistor shall be tested for all parameters listed.

4/ This is a go-no-go test, the limit values are used for input voltages.

5/ Each output shall be tested for all parameters listed. V_C = 20 V unless otherwise specified.

6/ A 2.7-kilohm pullup resistor is added from the Sync pin to V_{REF} to limit the effects of stray capacitance in automatic test equipment.

7/ Each output shall be tested for all parameters listed V_C = 15 V unless otherwise specified.

8/ Only use the shutdown pin, not the Sync pin, to deactivate the device.

sharp metallic corners.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional block diagrams. The functional block diagrams shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3 herein.

3.2.5 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 52 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004) 1/	1,2,3,4,5,6	1,2,3,4,5,6
Group A test requirements (method 5005)	1,2,3,4,5,6, 7,9,10,11	1,2,3,4,5, 6,7,9
Group C end-point and group B, class S electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits
Additional electrical subgroup for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1

1/ PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test, method 1015 of MIL-STD-883.

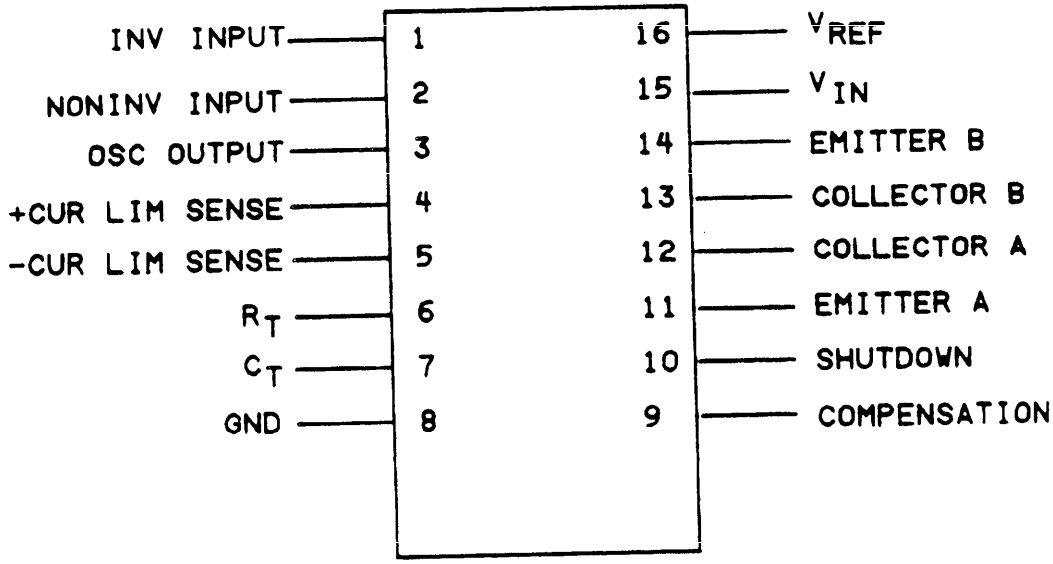
(1) For class S devices: Test condition D using the circuit shown on figure 3, and reverse bias burn-in (method 1015 of MIL-STD-883) using the circuit shown on figure 4.

(2) For class B devices: Test condition D using the circuit shown on figure 3.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameter tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) - The PDA for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

Device type 01



Device types 02 and 04

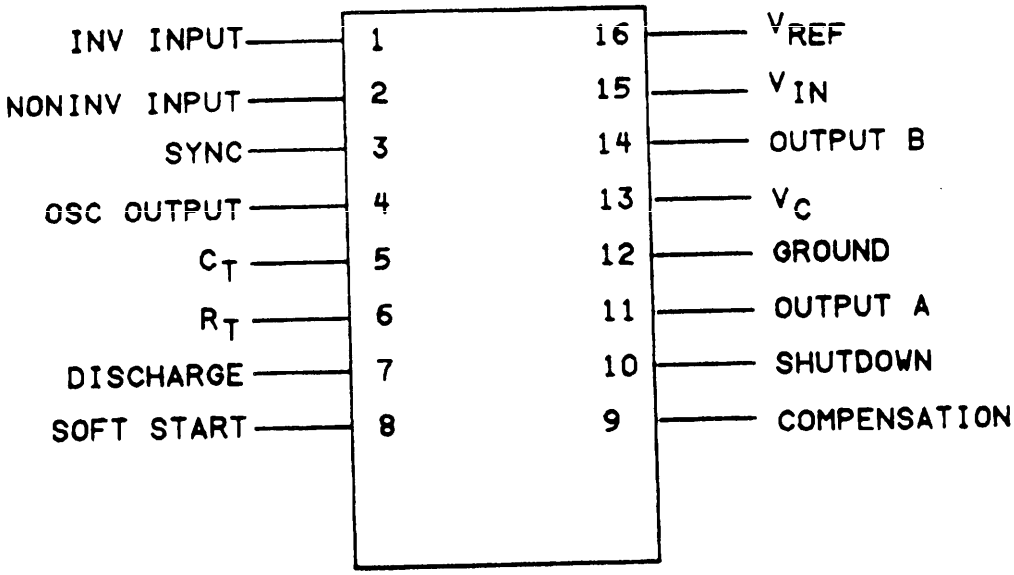


FIGURE 1. Terminal connections.

Device type 03

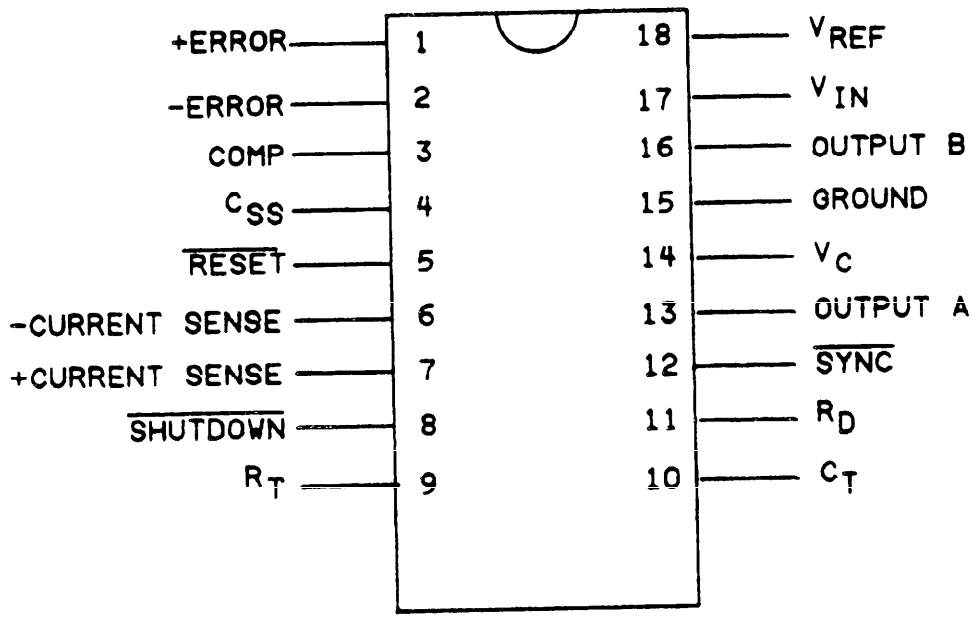


FIGURE 1. Terminal connections - Continued.

Device type 01

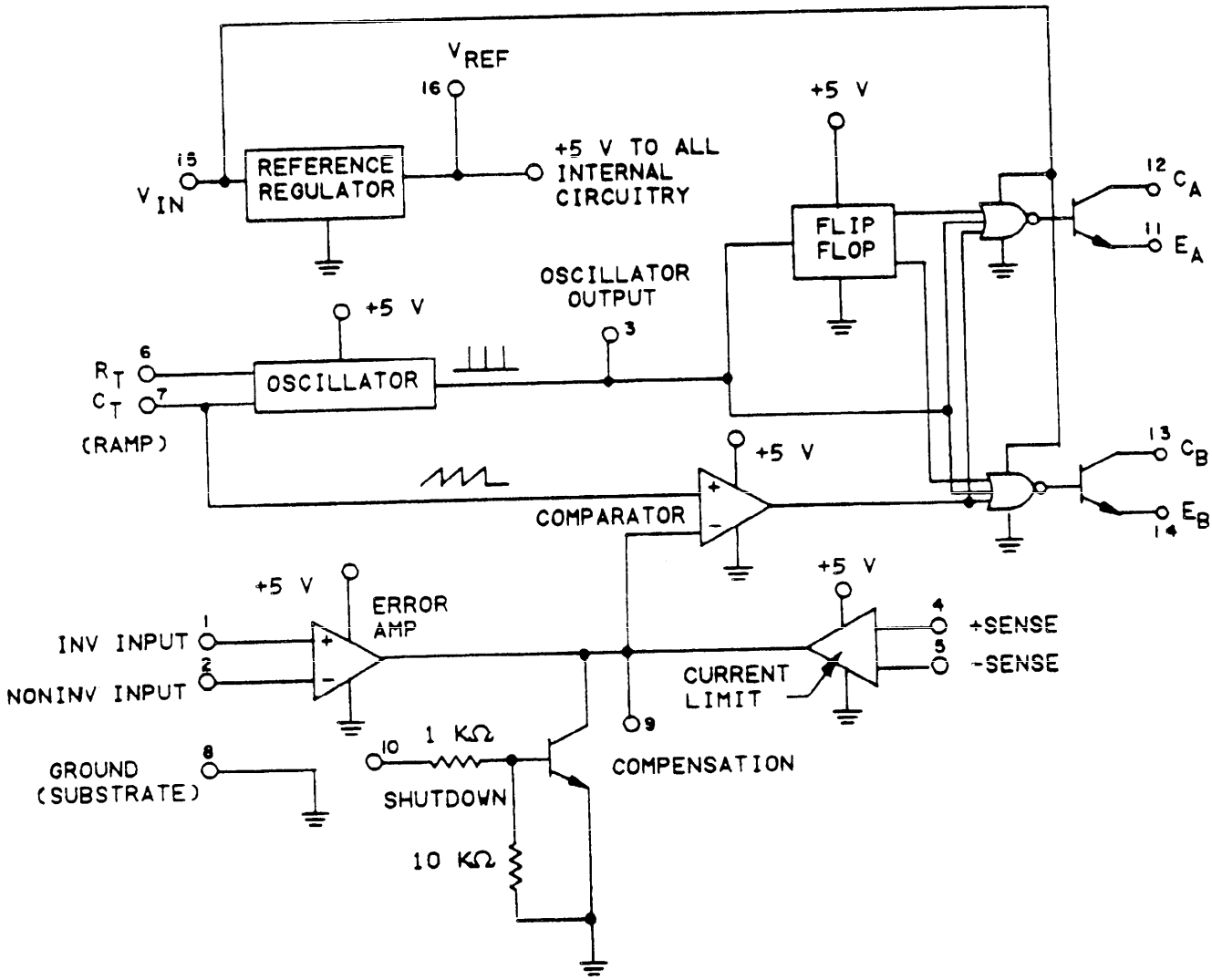


FIGURE 2. Functional block diagram.

Device types 02 and 04

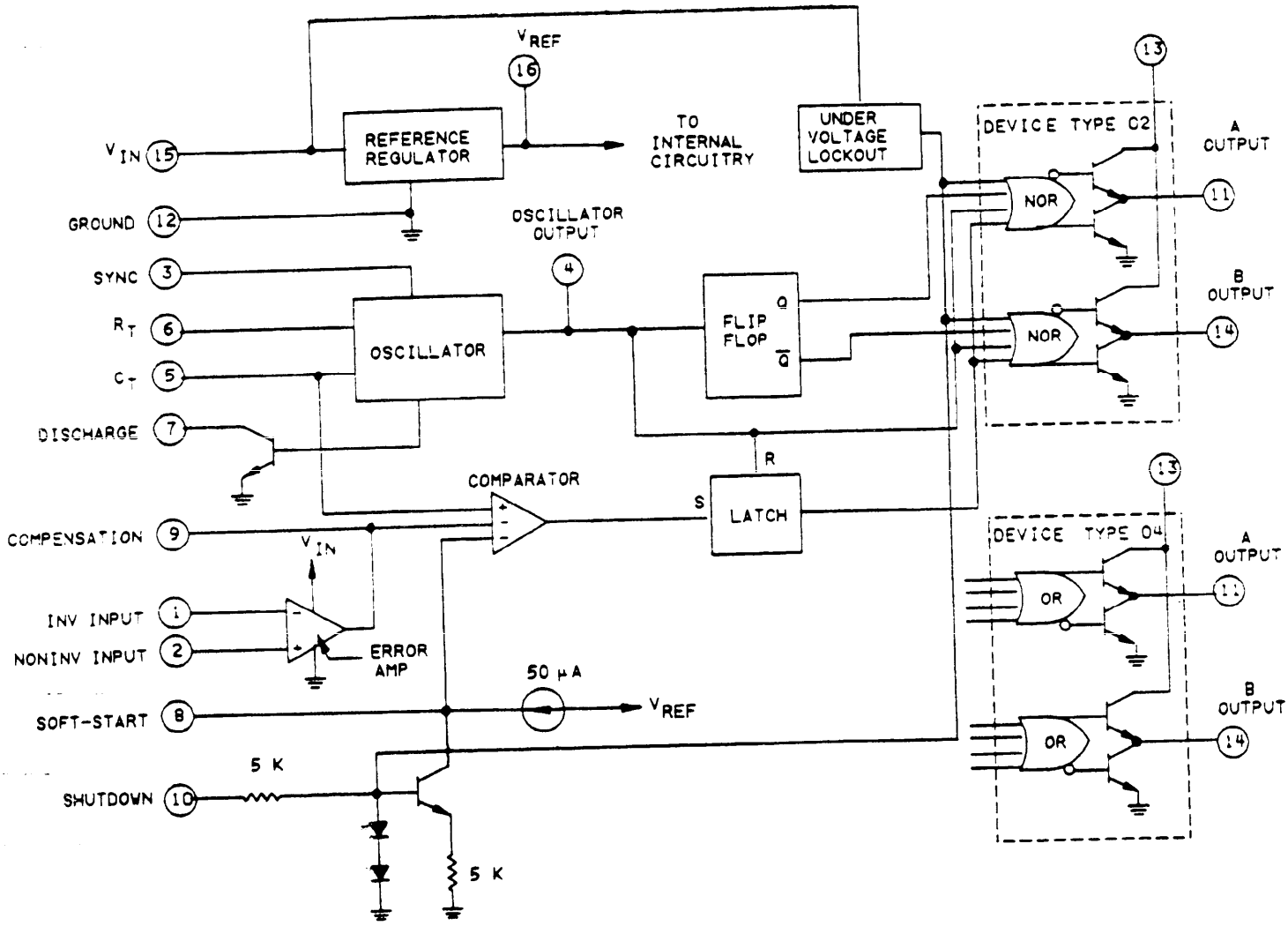


FIGURE 2. Functional block diagram - Continued.

Device type 03

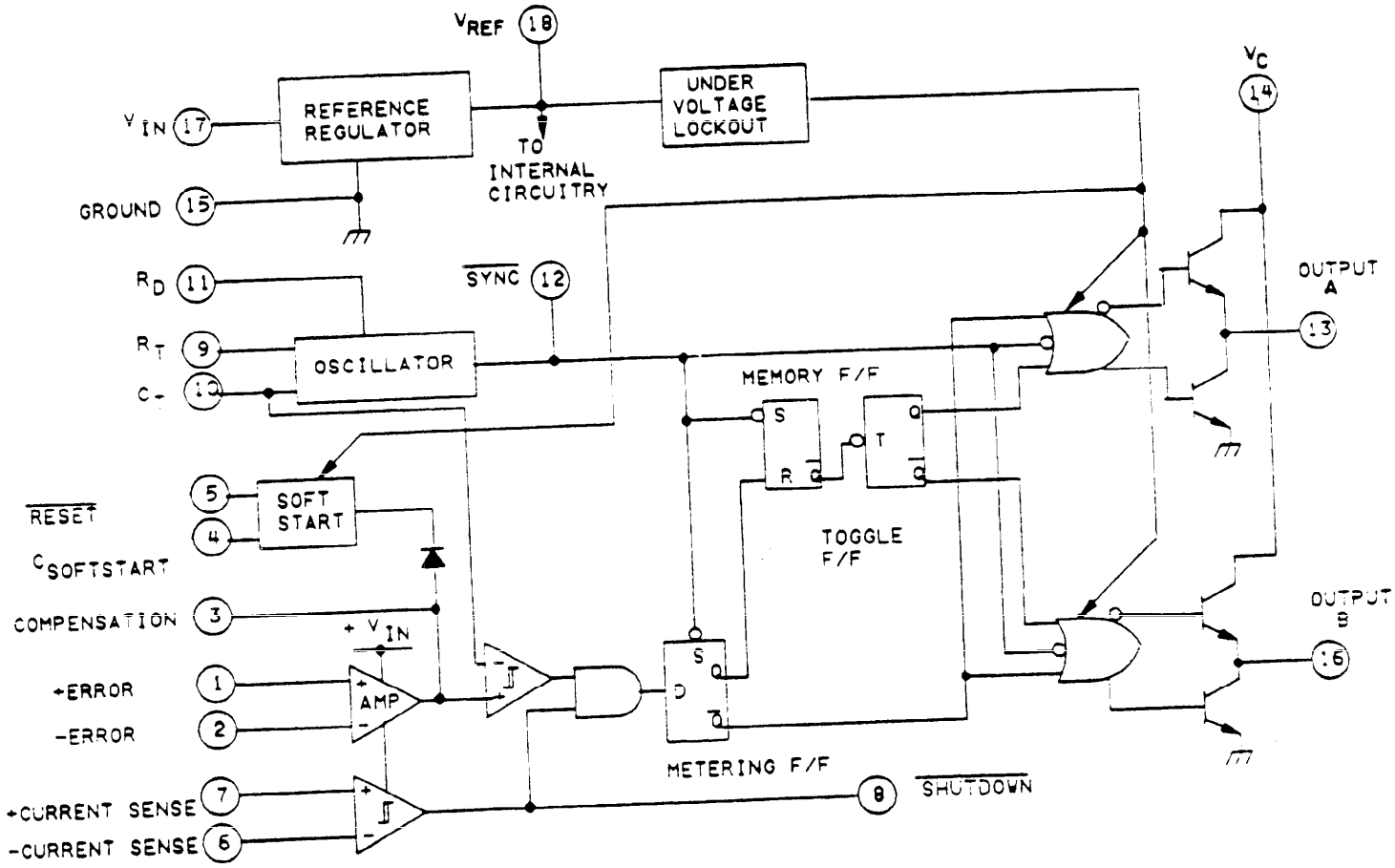
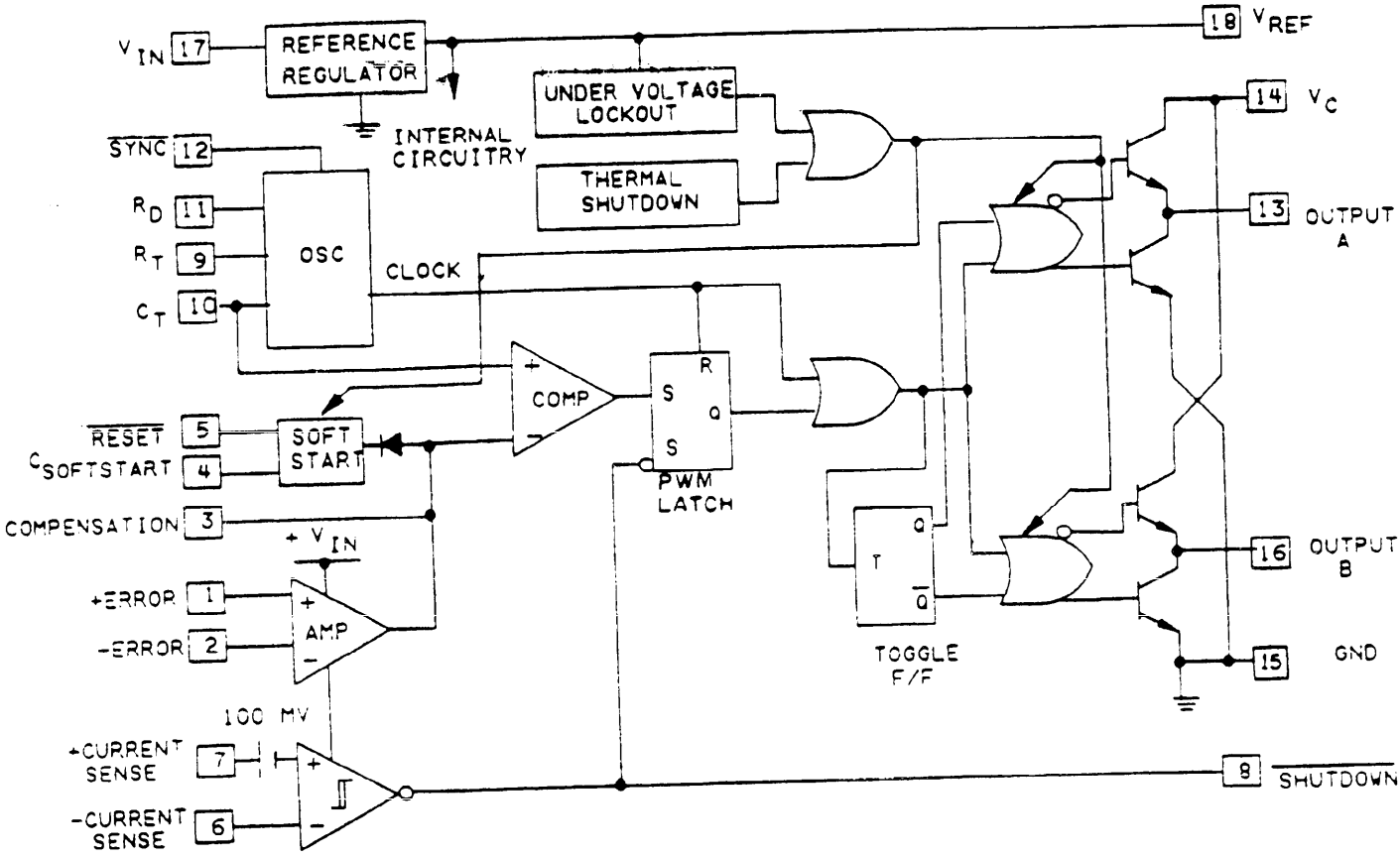


FIGURE 2. Functional block diagram - Continued.

Device type 03

(Unitrode)



NOTE: The oscillator's $\overline{\text{sync}}$ (pin 12) is not connected to the output gates.

FIGURE 2. Functional block diagram - Continued.

Device type 01

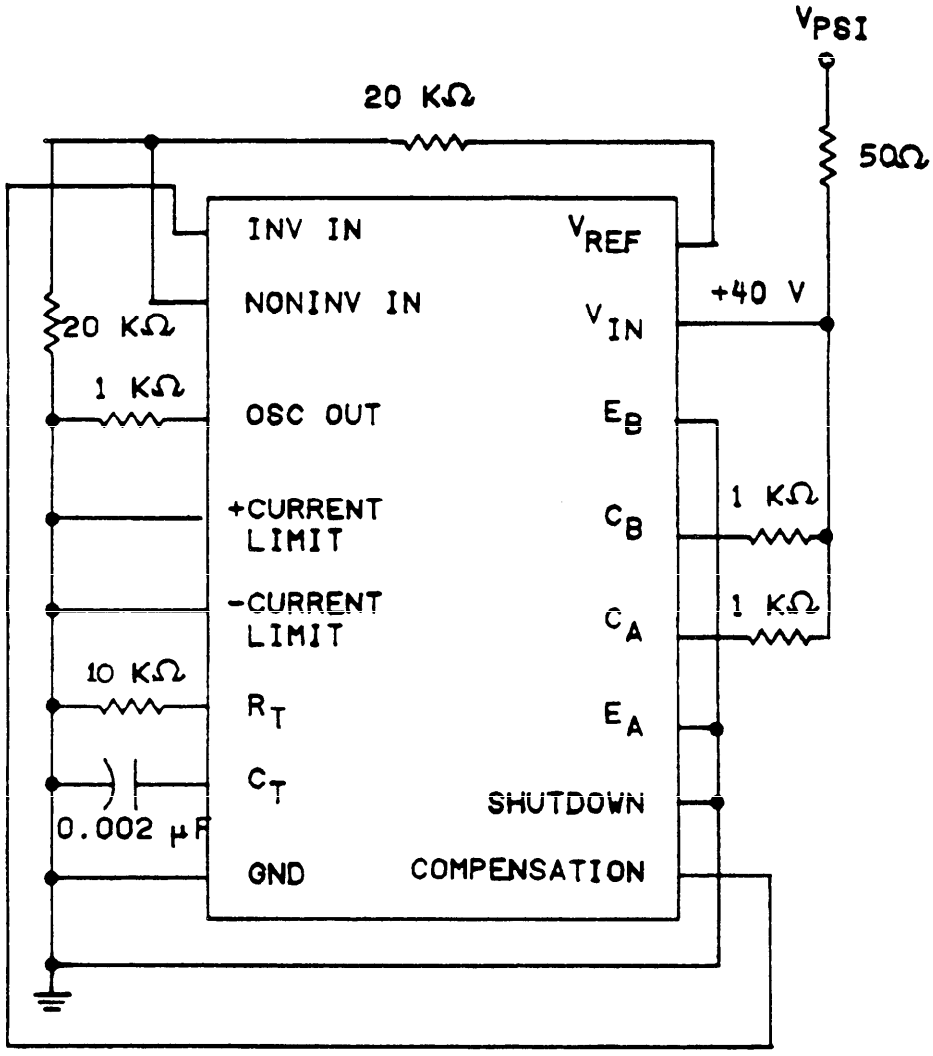


FIGURE 3. Test circuit burn-in and operating life test.

Device types 02 and 04

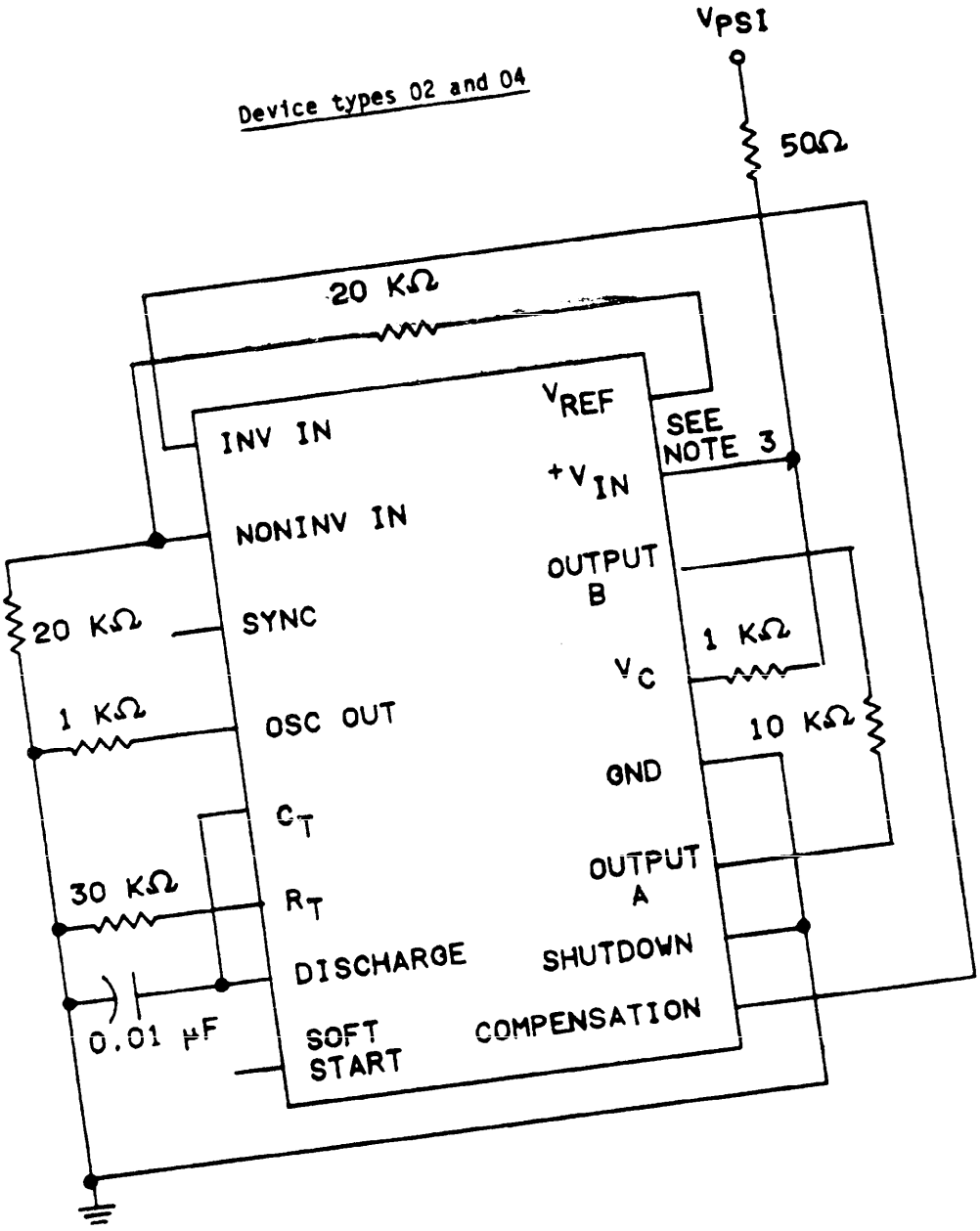
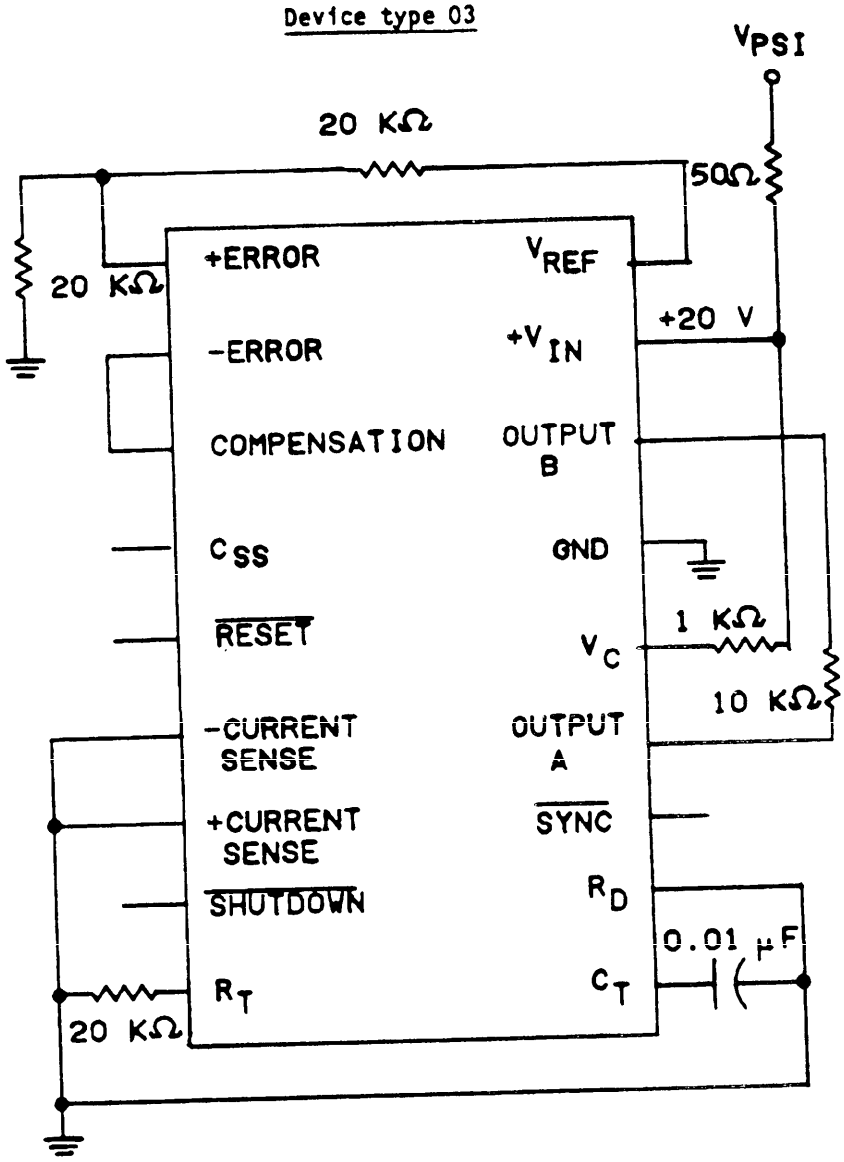


FIGURE 3. Test circuit burn-in and operating life test - Continued.

Device type 03



NOTES:
1. All resistors and capacitors are ±5%.
2. Bypass capacitors and compensation capacitors are allowed as needed to prevent oscillations.
3. V_{IN} = 30 V for class B, V_{IN} = 27 V for class S.

FIGURE 3. Test circuit burn-in and operating life test - Continued.

Device type 01

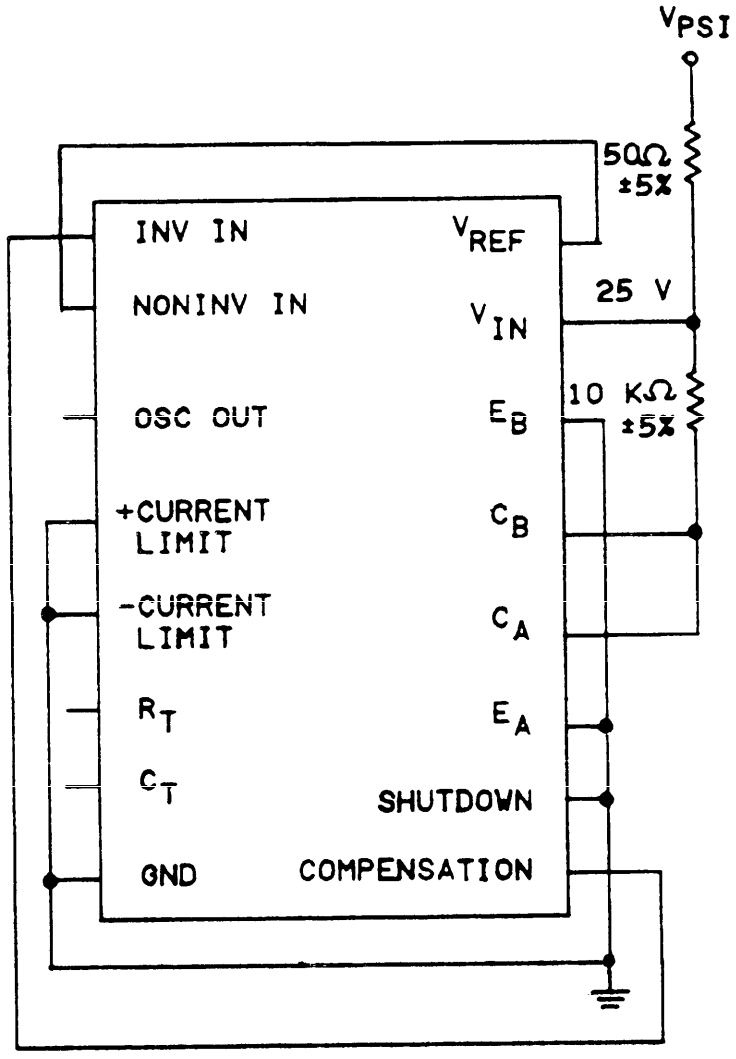


FIGURE 4. Test circuit, burn-in (steady state power and steady state life test.

Device types 02 and 04

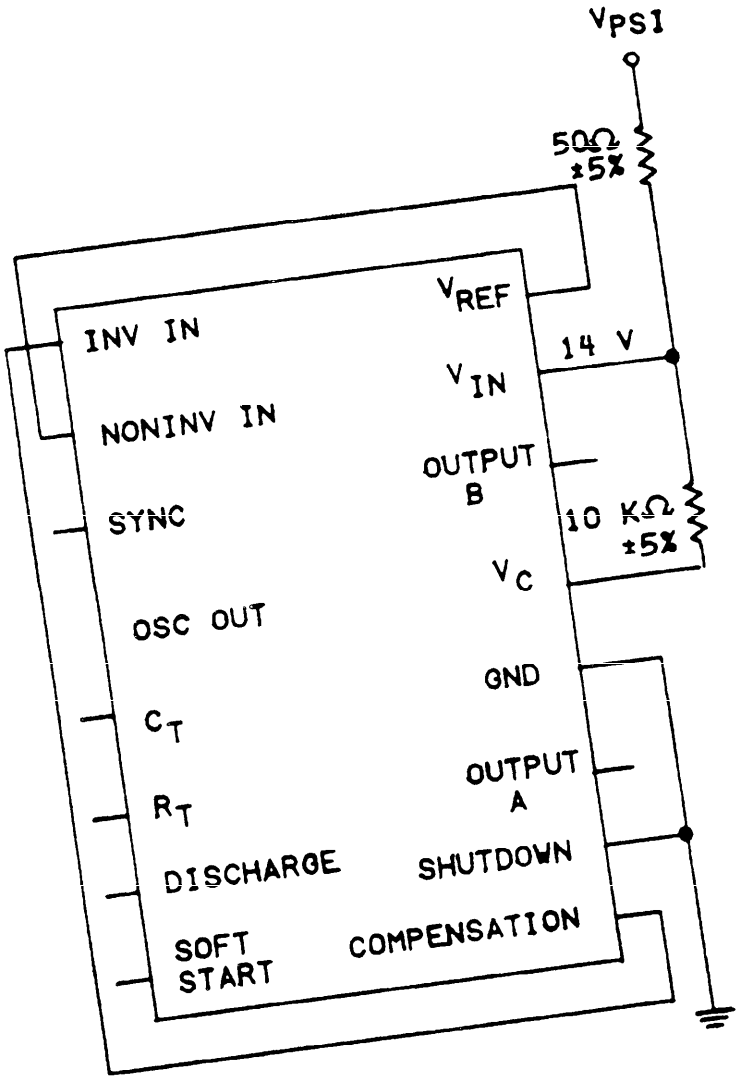
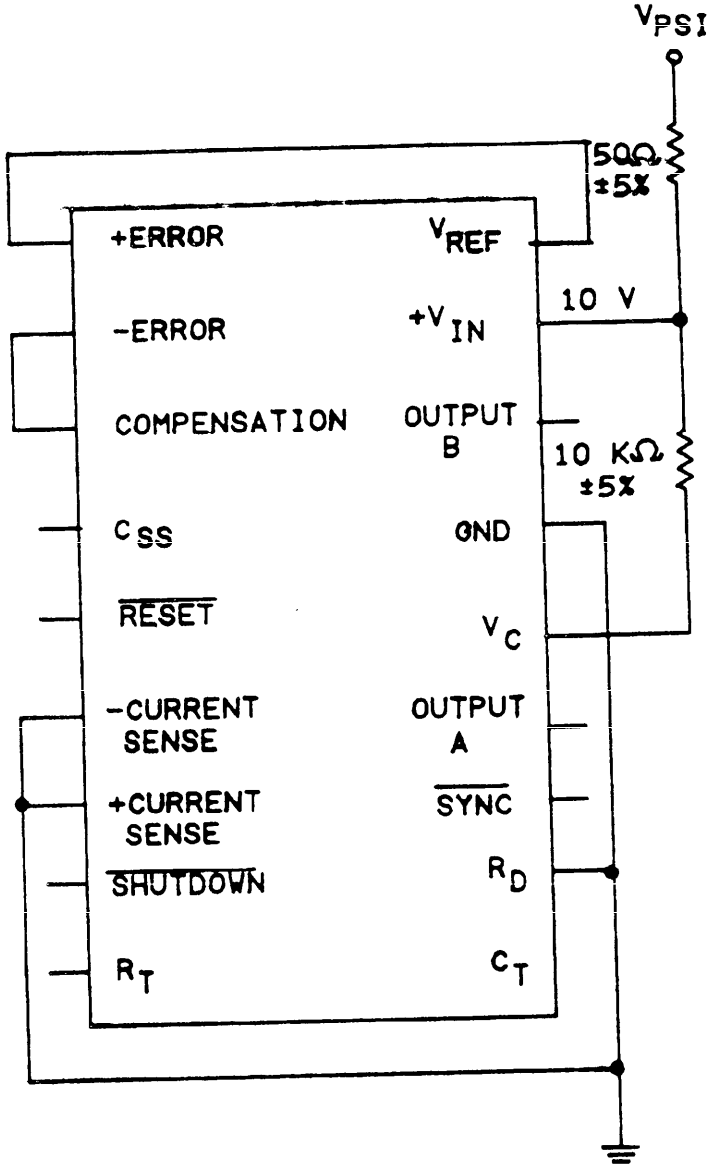


FIGURE 4. Test circuit, burn-in (steady state power and steady state life test - Continued.

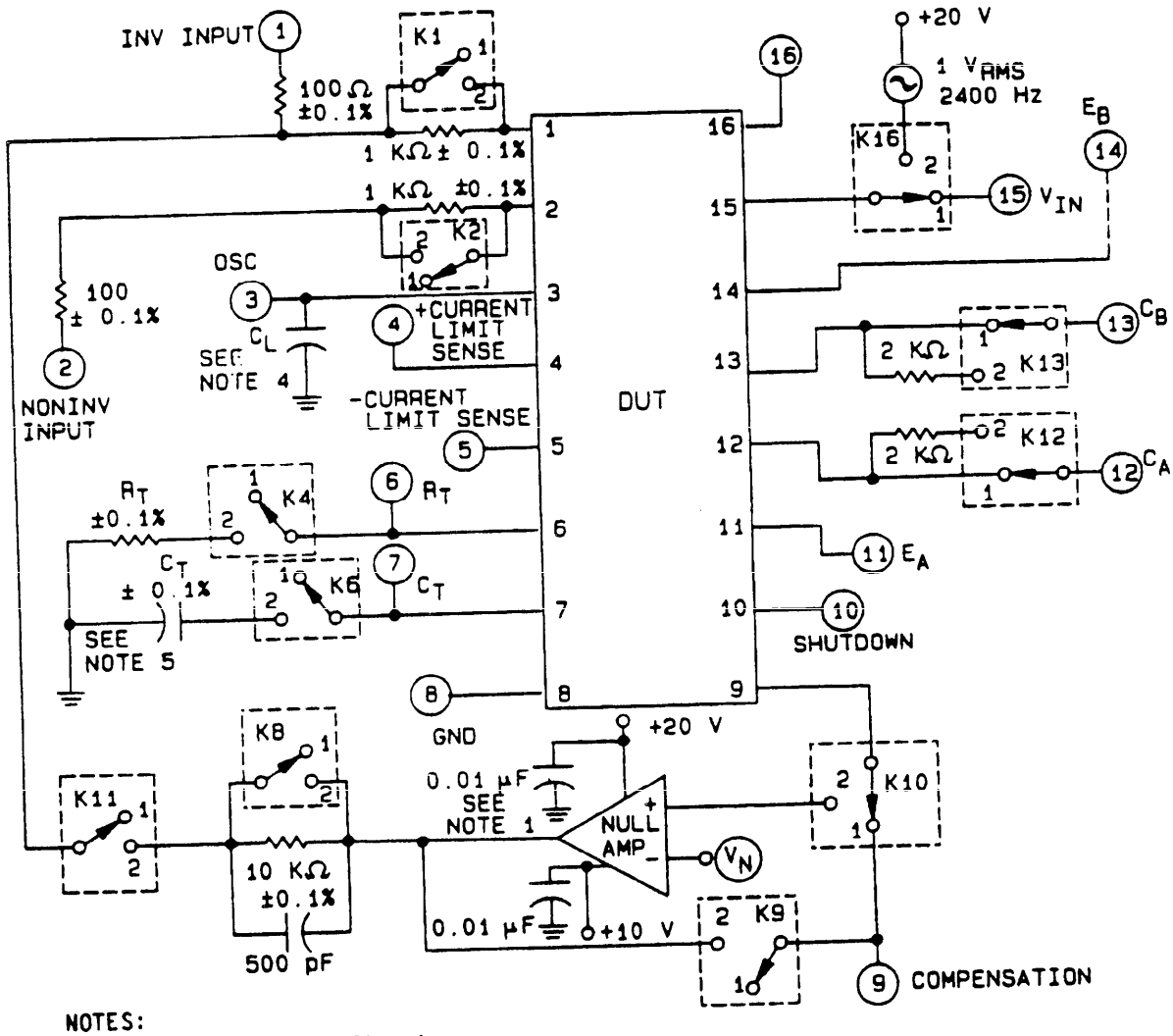
Device type 03



NOTE: Bypass and compensation capacitor are allowed to prevent oscillations.

FIGURE 4. Test circuit, burn-in (steady state power and steady state life test - Continued.

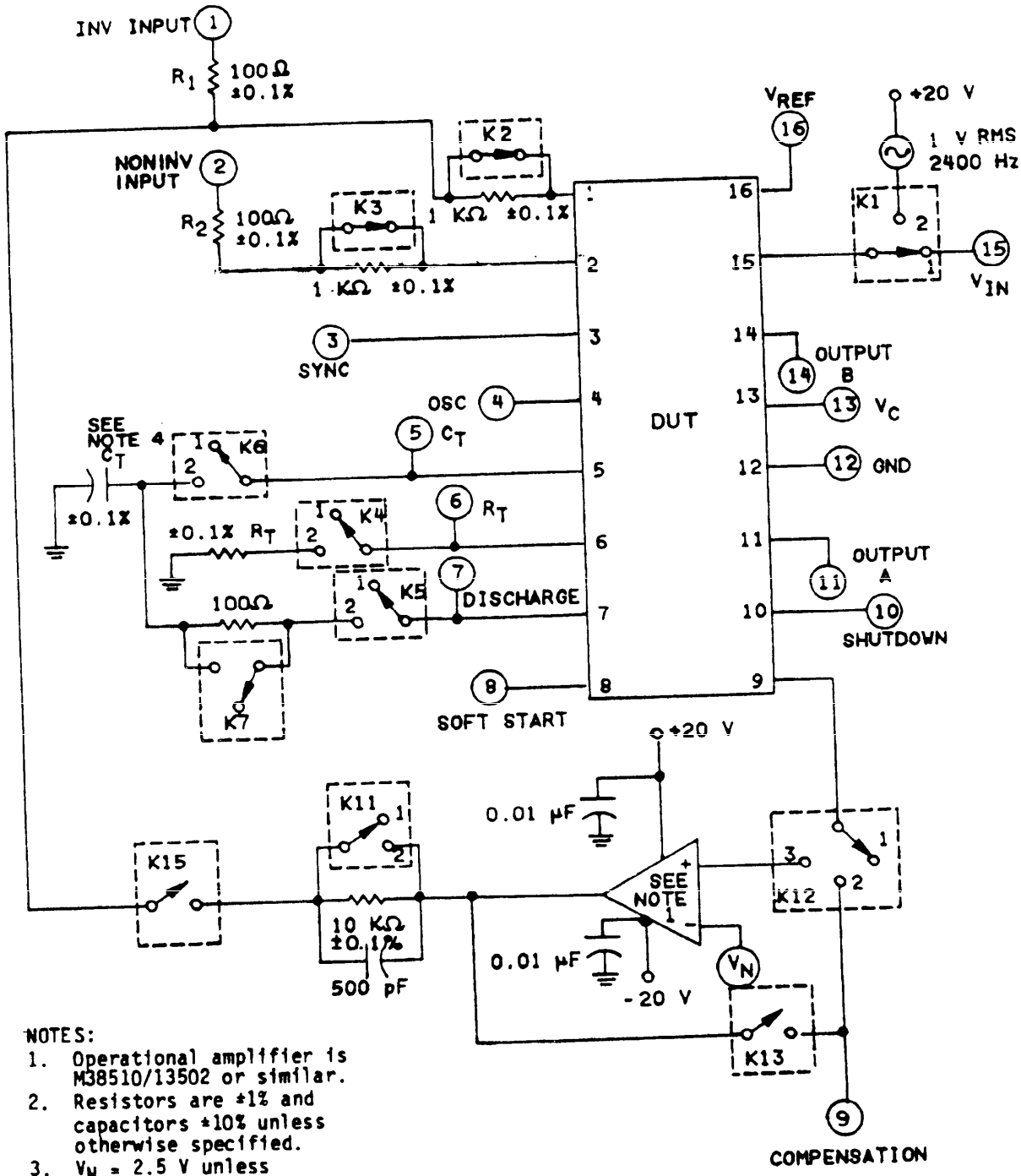
Device type 01



NOTES:
1. Operational amplifier is M38510/13502 or similar.
2. Resistors and are ±1% capacitors ±10% unless otherwise specified.
3. V_N = 2.5 V unless otherwise specified in table III.
4. C_L ≤ 20 pF (including probe and jig capacitance).
5. ±1.0% for t_{OSC} (max) test.

FIGURE 5. Test circuit for static and dynamic tests.

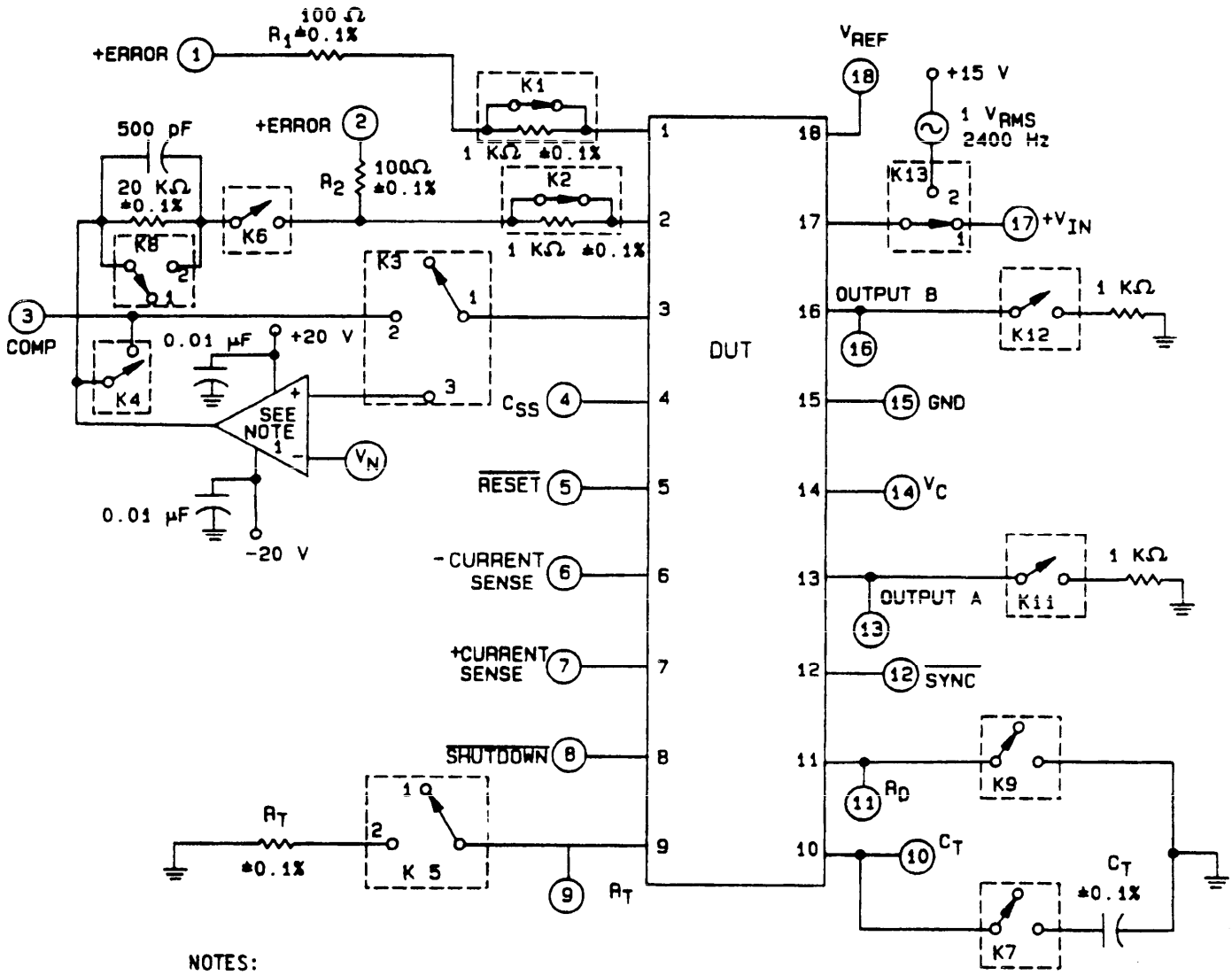
Device types 02 and 04



NOTES:
1. Operational amplifier is M38510/13502 or similar.
2. Resistors are ±1% and capacitors ±10% unless otherwise specified.
3. V_N = 2.5 V unless otherwise specified in table III.
4. ±1.0% for f_{OSC} (min) and f_{OSC} (max) tests.

FIGURE 5. Test circuit for static and dynamic tests - Continued.

Device type 03



NOTES:
1. Operational amplifier is M38510/13502 or similar.
2. Resistors are ±1% and capacitors ±10% unless otherwise specified.
3. V_N = 2.5 V unless otherwise specified in table III.
4. ±1.0% for f_{OSC} (min) and f_{OSC} (max) tests.

FIGURE 5. Test circuit for static and dynamic tests - Continued.

Device type 01

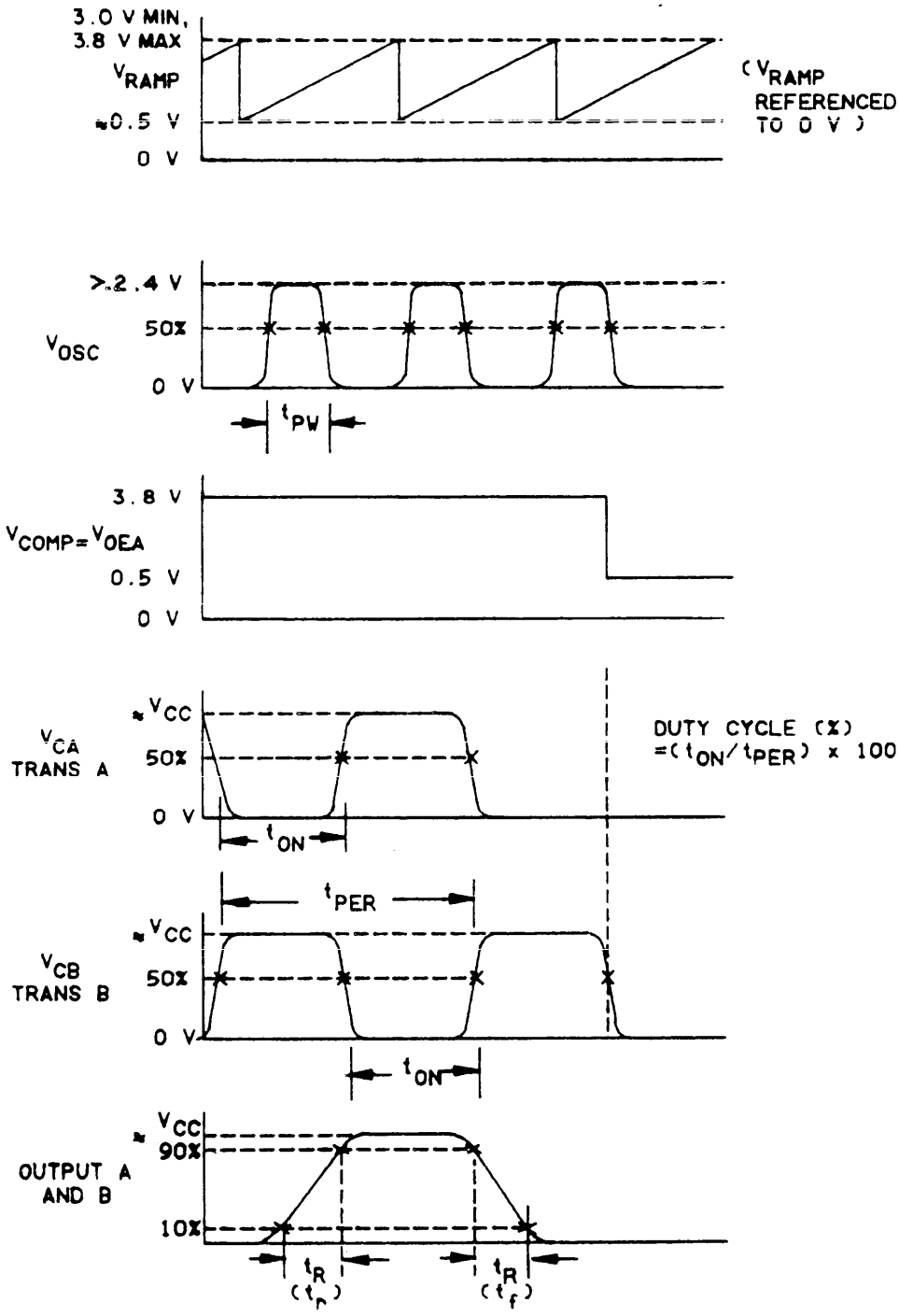
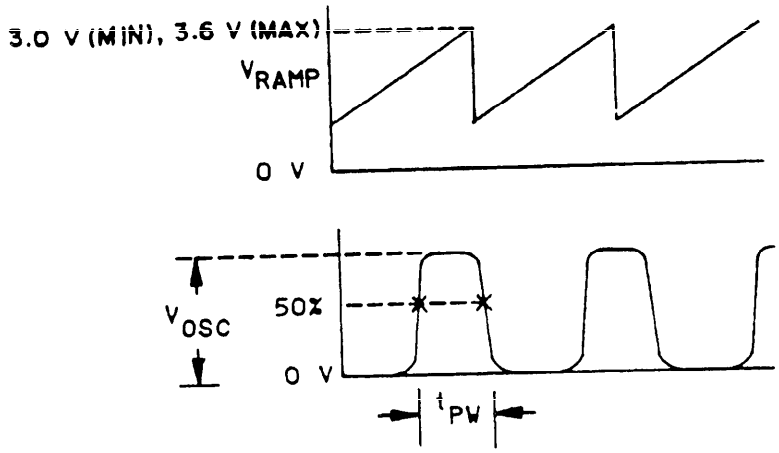
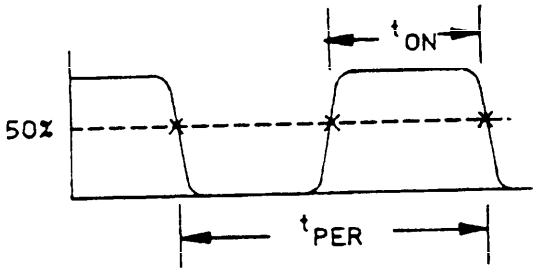


FIGURE 6. Test waveforms.

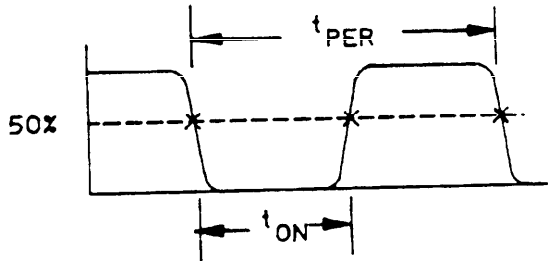
Device types 02 and 04



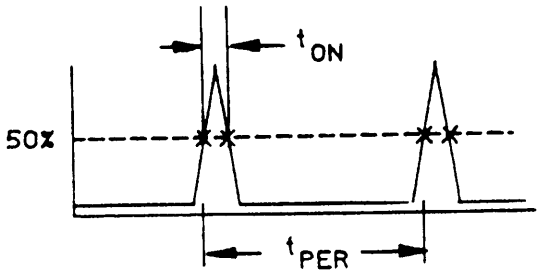
Device type 02

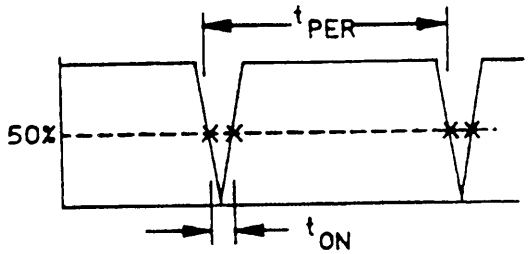


Device type 04



a. Maximum duty cycle

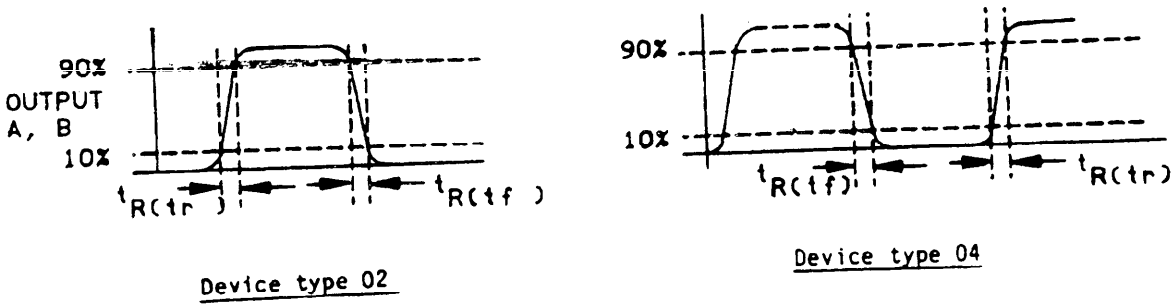




b. Approximate minimum duty cycle (actual = 0%)
Duty cycle (%) = (t_{ON}/t_{PER}) x 100

FIGURE 6. Test waveforms - Continued

Device types 02 and 04



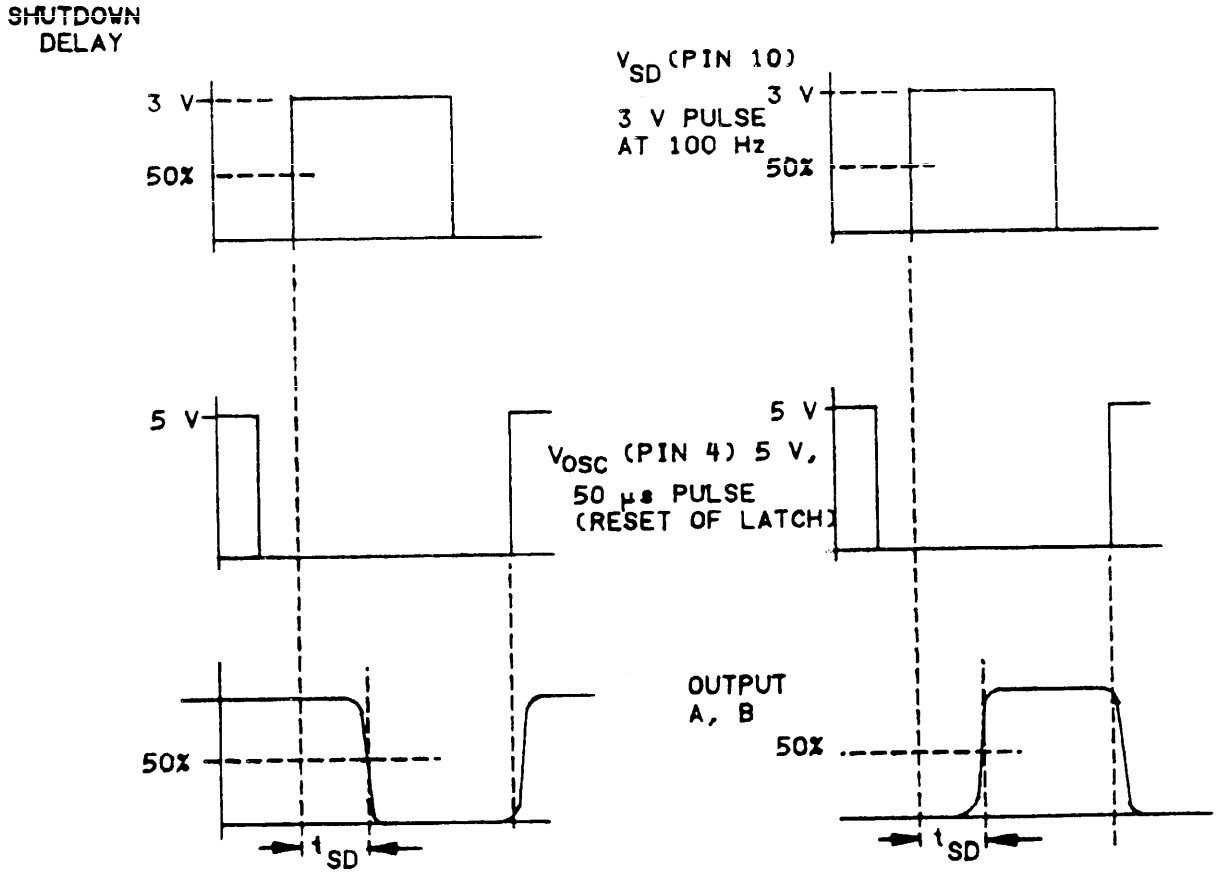


FIGURE 6. Test waveforms - Continued

Device types 02 and 04

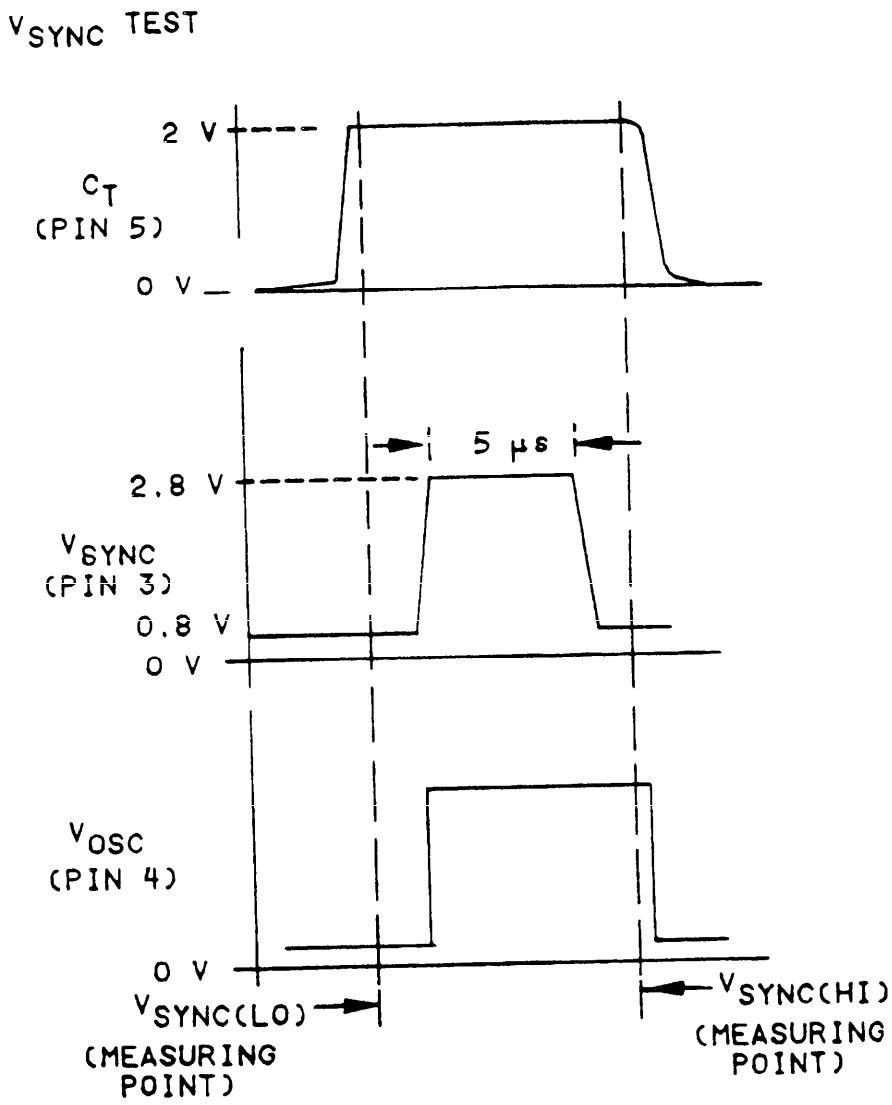


FIGURE 6. Test waveforms - Continued

Device type 03

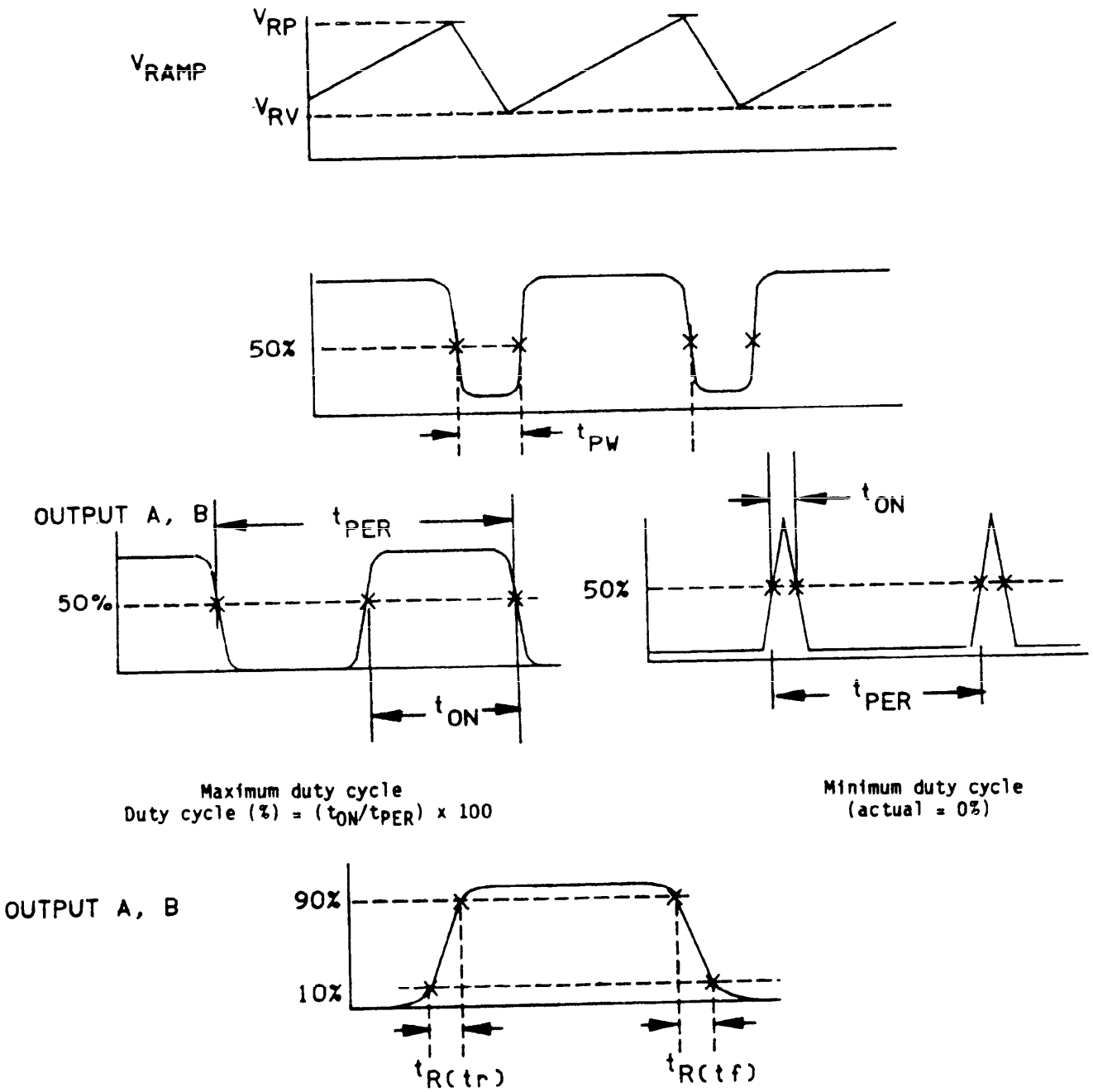


FIGURE 6. Test waveforms - Continued.

Device type 03

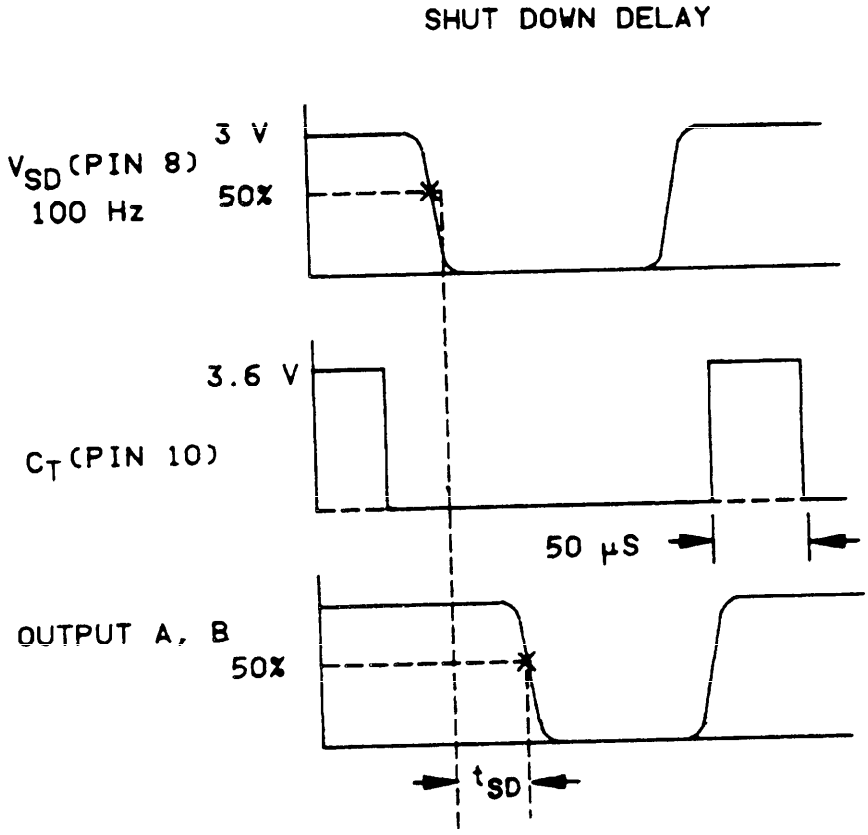
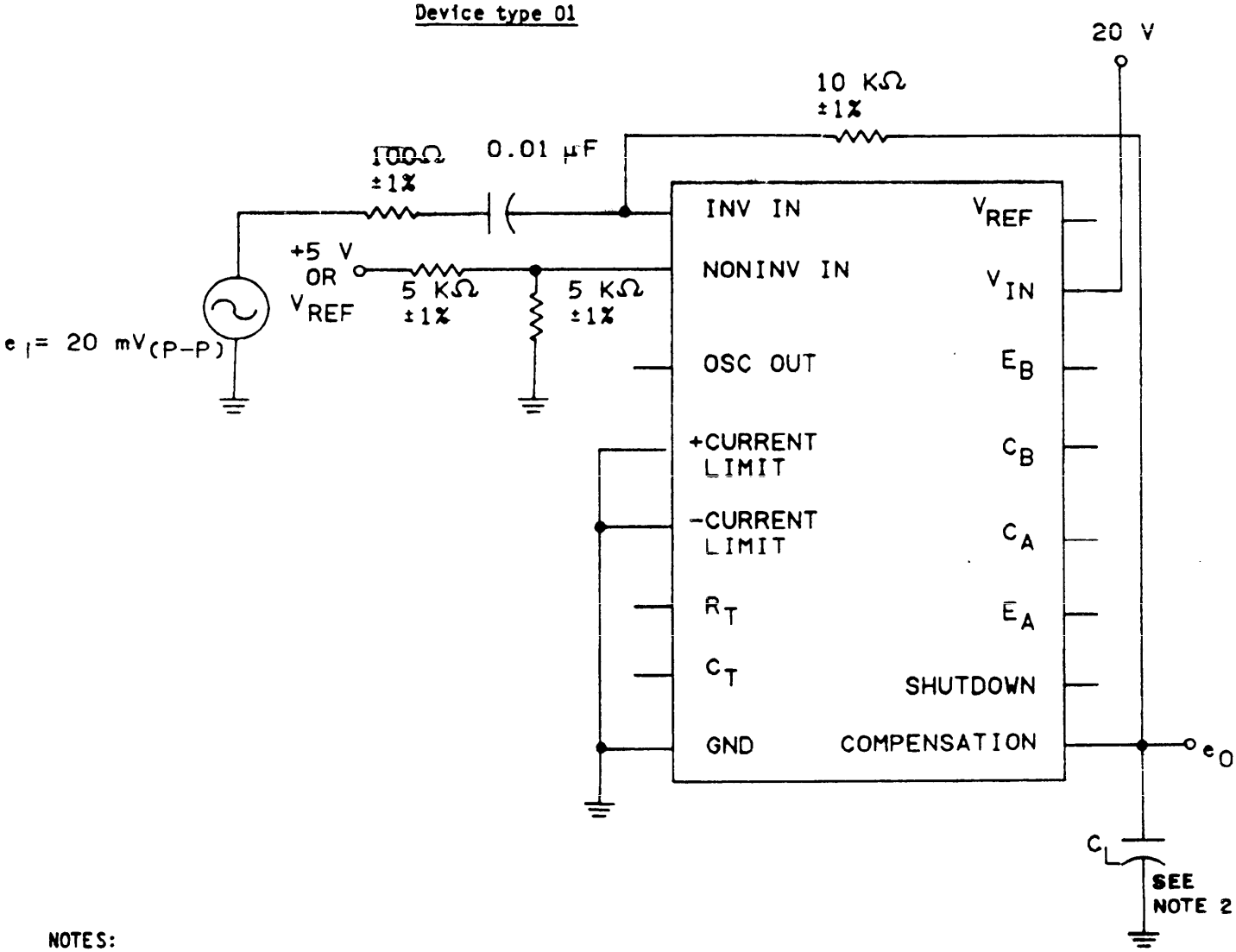


FIGURE 6. Test waveforms - Continued.

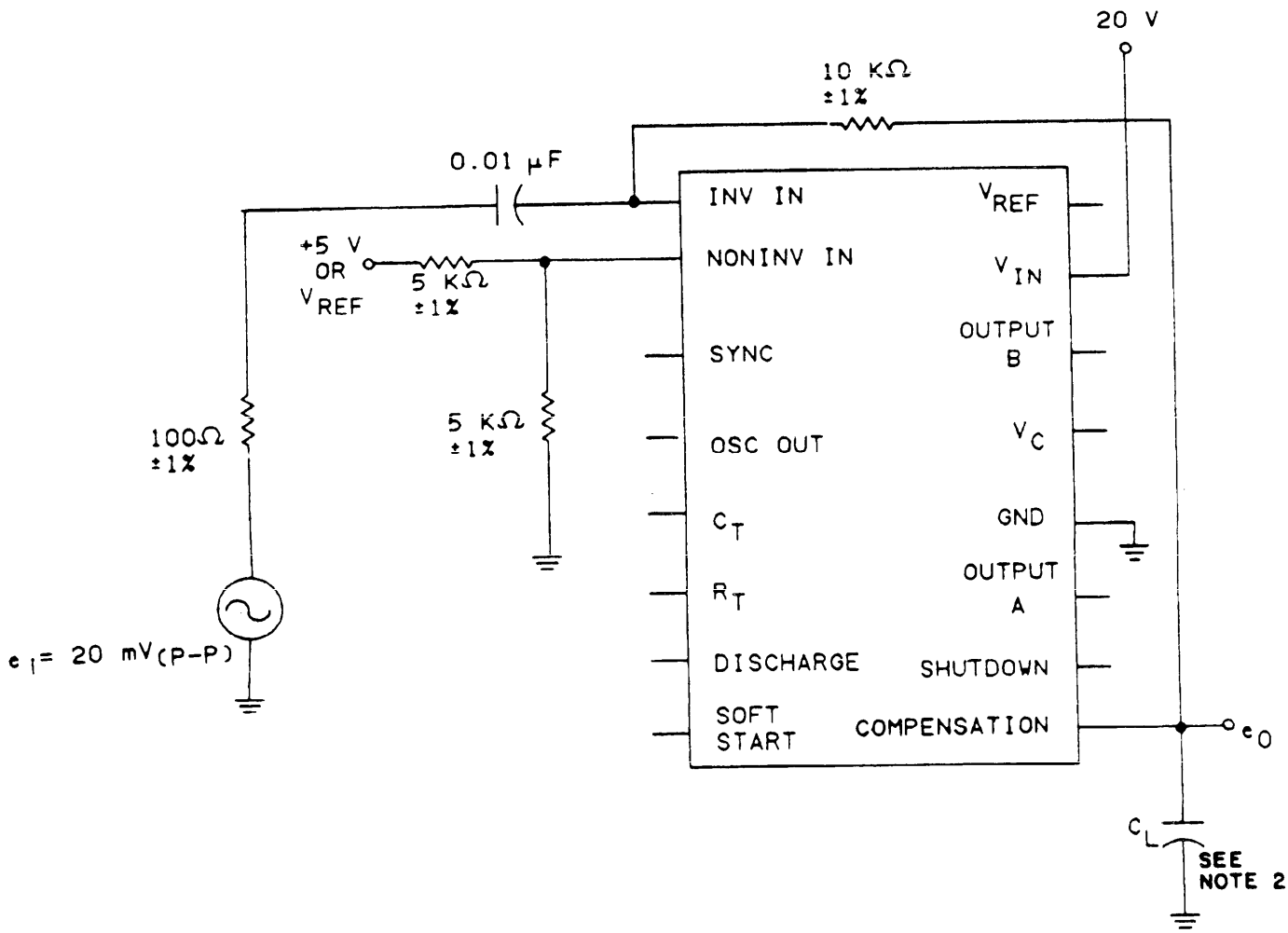
Device type 01



NOTES:
1. G_{BW} in measured by increasing signal frequency (starting at 100 kHz) until e_O = 20 $mV_{(P-P)}$.
   The frequency at which this occurs is G_{BW}.
2. (C_L) capacitance load = 20 pF, -10%, on e_O including scope probe and jig capacitance.
3. Alternate method: Set signal frequency (e_I) to the G_{BW} minimum limit if $e_O \geq$ 20 $mV_{(P-P)}$ then G_{BW} is $\geq$ the minimum limit.

FIGURE 7. Unity-gain bandwidth test circuit.

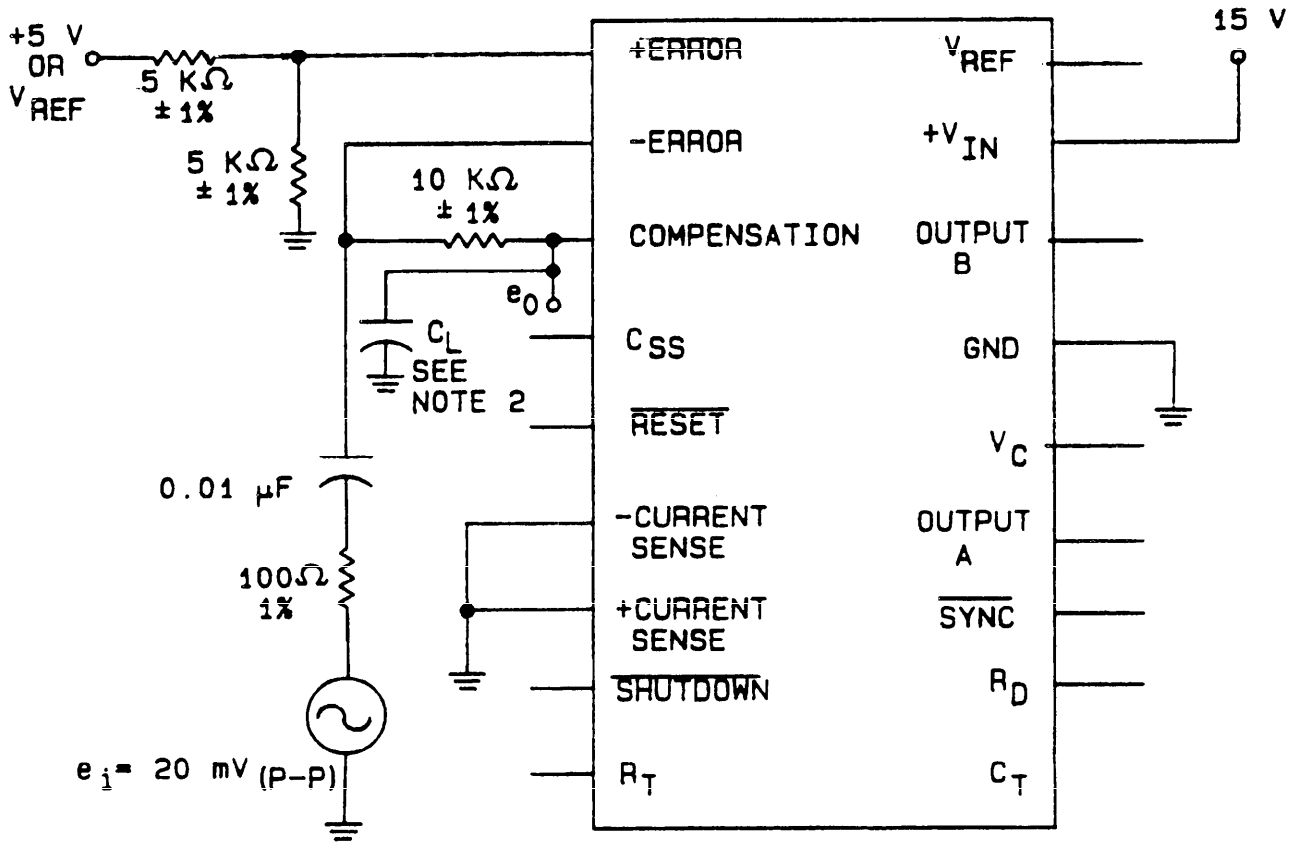
Device types 02 and 04



NOTES:
1. G_{BW} in measured by increasing signal frequency (starting at 100 kHz) until e_O = 20 $mV_{(P-P)}$.
   The frequency at which this occurs is G_{BW}.
2. (C_L) capacitance load = 20 pF, -10%, on e_O including scope probe and jig capacitance.
3. Alternate method: Set signal frequency (e_i) to the G_{BW} minimum limit if $e_O \geq$ 20 $mV_{(P-P)}$ then G_{BW} is $\geq$ the minimum limit.

FIGURE 7. Unity-gain bandwidth test circuit - Continued.

Device type 03



NOTES:
1. G_{BW} in measured by increasing signal frequency (starting at 100 kHz) until e_0 = 20 $mV_{(P-P)}$.
   The frequency at which this occurs is G_{BW}.
2. (C_L) capacitance load = 20 pF, -10%, on e_0 including scope probe and jig capacitance.
3. Alternate method: Set signal frequency (e_i) to the G_{BW} minimum limit if $e_0 \geq$ 20 $mV_{(P-P)}$ then G_{BW} is $\geq$ the minimum limit.

FIGURE 7. Unity-gain bandwidth test circuit - Continued.

TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	Adapter pin numbers															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			INV INPUT	NONINV INPUT	OSC	+CL	-CL	R_T	C_T	GND	COMP	SHUT-DOWN	E_A	C_A	C_B	E_B	V_{IN}	V_{REF}
T_C = +25°C 1/	V_{IO}	1	2.5 V	2.5 V		GND	GND			GND							20 V	
	$+I_{IB}$	2	"	"		"	"			"							"	
	$-I_{IB}$	3	"	"		"	"			"							"	
	I_{COS1}	4	2.575 V	2.425 V		"	"			"	I_1 to 2.5 V						"	
	I_{COS2}	5	2.425 V	2.575 V		"	"			"	I_2 to 2.5 V						"	
	I_{IO}	6																
	C_{MR}	7	1.8 V	1.8 V		GND	GND			GND							20 V	
			3.4 V	3.4 V		"	"			"							"	
	V_{REF}	8	2.5 V	2.5 V		"	"			"							"	
	V_{RLIME}	9	"	"		"	"			"							8 V	
			"	"		"	"			"							40 V	
	V_{RLOAD}	10	"	"		"	"			"							20 V	-20 mA
	I_{OS}	11	"	"		"	"			"							"	0 V
	$V_{SAT(OSC)}$	12	"	"	5 V	"	"		5 mA	"							"	
	$I_{CEX(A)}$	13	"	"		"	"			"		1.4 V	GND	40 V			"	
	$I_{CEX(B)}$	14	"	"		"	"			"		1.4 V			40 V	GND	"	
	$V_{SAT(A)}$	15	2.425 V	2.575 V	1/	"	"			"			GND	50 mA			"	
	$V_{SAT(B)}$	16	"	"	"	"	"			"					150 mA	GND	"	
	$V_{EO(A)}$	17	"	"	"	"	"			"			-50 mA	20 V			"	
	$V_{EO(B)}$	18	"	"	"	"	"			"					120 V	-50 mA	"	
	I_{IN}	19	2.5 V	2.5 V		"	"			"							40 V	

See footnotes at end of table.

TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
1 T_C = +25°C	V_{IO}	1	K1, K2, K9, K10, K11	9	E1	V	$V_{IO} = \frac{(E1-2.5\ V)}{101}$	-5	5	mV
	$+I_{IB}$	2	K1, K9, K10, K11	"	E2	"	$+I_{IB} = \frac{E1-E2}{101000}$	0.01	10	µA
	$-I_{IB}$	3	K2, K9, K10, K11	"	E3	"	$-I_{IB} = \frac{E3-E1}{101000}$	0.01	10	µA
	I_{COS1}	4		"	I1	µA	$I_{COS1} = I1$	65	170	µA
	I_{COS2}	5		"	I2	µA	$I_{COS2} = I2$	-170	-65	µA
	I_{IO}	6					$I_{IO} = (+I_{IB})-(-I_{IB})$	-1	1	µA
	C_{MR}	7	K1, K2, K9, K10, K11	9 9	E4 E5	V "	$CMR = 20\ Log \left[\frac{(3.4 - 1.8)\ 101}{\lvert E5-E4-1.6\rvert}\right]$	70		dB
	V_{REF}	8		16	E6	"	$V_{REF} = E6$	4.8	5.2	V
	V_{RLINE}	9		" "	E7 E8	" "	$V_{RLINE} = E8 - E7$	-20	20	mV
	V_{RLOAD}	10		"	E9	"	$V_{RLOAD} = E9 - E6$	-50	50	mV
	I_{OS}	11		"	I3	mA	$I_{OS} = I3$	-190		mA
	$V_{SAT(OSC)}$	12		7	E10	V	$V_{SAT(OSC)} = E10$	0.7	1.1	V
	$I_{CEX(A)}$	13		12	I4	µA	$I_{CEX(A)} = I4$		10	µA
	$I_{CEX(B)}$	14		13	I5	µA	$I_{CEX(B)} = I5$		10	µA
	$V_{SAT(A)}$	15		12	E11	V	$V_{SAT(A)} = E11$		2.0	V
	$V_{SAT(B)}$	16		13	E12	"	$V_{SAT(B)} = E12$		2.0	"
	$V_{EO(A)}$	17		11	E13	"	$V_{EO(A)} = E13$	17		"
	$V_{EO(B)}$	18		14	E14	"	$V_{EO(B)} = E14$	17		"
	I_{IN}	19		15	I6	mA	$I_{IN} = I6$		10	mA

See footnotes at end of table.

TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			INV INPUT	NONINV INPUT	OSC	$^{+}C_L$	$^{-}C_L$	R_T	C_T	GND	COMP	SHUT-DOWN	E_A	C_A	C_B	E_B	V_{IN}	V_{REF}
2 T_C = +125°C	V_{IO}	20	2.5 V	2.5 V		GND	GND			GND							20 V	
	$^{+}I_{IB}$	21	"	"		"	"			"							"	
	$^{-}I_{IB}$	22	"	"		"	"			"							"	
	I_{IO}	23																
	C_{MR}	24	1.8 V	1.8 V		GND	GND			GND							20 V	
			3.4 V	3.4 V		"	"			"							"	
	V_{REF}	25	2.5 V	2.5 V		"	"			"							"	
	V_{RLINE}	26	"	"		"	"			"							6 V	
			"	"		"	"			"							40 V	
	V_{RLOAD}	27	"	"		"	"			"							20 V	-20 mV
	I_{OS}	28	"	"		"	"			"							"	0 V
	$I_{CEX(A)}$	29	"	"		"	"			"		1.4 V	GND	40 V			"	
	$I_{CEX(B)}$	30	"	"		"	"			"		1.4 V			40 V	GND	"	
	$V_{SAT(A)}$	31	2.425 V	2.575 V	1/	"	"			"			GND	50 mA			"	
	$V_{SAT(B)}$	32	2.425 V	2.575 V	"	"	"			"					50 mA	GND	"	
	I_{IN}	33	2.5 V	2.5 V		"	"			"							40 V	
3 T_C = -55°C		34 to 47	(All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 2, T_C = +125°C. Except for I_{OS} at -55°C. The minimum limit is -190 mA).															

See footnotes at end of table.

TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
2 T_C = +125°C	V_{IO}	20	K1, K2, K9, K10, K11	9	E15	V	$V_{IO} = \frac{(E15-2.5\ V)}{101}$	-5	5	mV
	$+I_{IB}$	21	K1, K9, K10, K11	"	E16	"	$+I_{IB} = \frac{E15-E16}{101000}$	0.01	10	µA
	$-I_{IB}$	22	K2, K9, K10, K11	"	E17	"	$-I_{IB} = \frac{E17-E15}{101000}$	0.01	10	µA
	I_{IO}	23					$I_{IO} = (+I_{IB})-(-I_{IB})$	-1	1	µA
	C_{MR}	24	K1, K2, K9, K10, K11	9 9	E18 E19	V "	$CMR = 20\ Log \left[\frac{(3.4 - 1.8)\ 101}{E19-E18-1.6}\right]$	70		dB
	V_{REF}	25		16	E20	"	$V_{REF} = E20$	4.8	5.2	V
	V_{RLINE}	26		" "	E21 E22	" "	$V_{RLINE} = E22-E21$	-20	20	mV
	V_{RLOAD}	27		"	E23	"	$V_{RLOAD} = E23-E20$	-50	50	mV
	I_{OS}	28		"	I7	mA	$I_{OS} = I7$	-120		mA
	$I_{CEX(A)}$	29		12	I8	µA	$I_{CEX(A)} = I8$		10	µA
	$I_{CEX(B)}$	30		13	I9	µA	$I_{CEX(B)} = I9$		10	µA
	$V_{SAT(A)}$	31		12	E24	V	$V_{SAT(A)} = E24$		2.0	V
	$V_{SAT(B)}$	32		13	E25	V	$V_{SAT(B)} = E25$		2.0	V
	I_{IN}	33		15	I10	mA	$I_{IN} = I10$		10	mA
3 T_C = -55°C		34 to 47								

See footnotes at end of table.

TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			INV INPUT	NONINV INPUT	OSC	R_{CL}	C_{CL}	R_T	C_T	GND	COMP	SHUT-DOWN	E_A	C_A	C_B	E_B	V_{IN}	V_{REF}
4/ T_C = +25°C	A_{VS}	48	2.5 V	2.5 V		GND	GND			GND							20 V	
			"	"		"	"			"							20 V	
	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$	49	"	"						"							20 V " 1 V rms	
	$f_{OSC(MAX)}$	50	"	"		GND	GND	2 kΩ	0.001 µF	"							20 V	
	f_{OSC1}	51	"	"		"	"	"	3/	"							20 V	
	$\frac{\Delta f_{OSC1}}{\Delta V_{IN}}$	52	"	"		"	"	"	.01 µF	"							8 V	
			"	"		"	"	"	"	"							40 V	
	V_{OSC}	53	2.5 V	2.5 V		"	"	"	"	"							20 V	
	V_{RAMP}	54	"	"		"	"	"	"	"							"	
	V_{SEN}	55	2.425 V	2.575 V			"	"	"	"							"	
	$V_{SD(HI)}$	56	"	"		GND	"		"	"		1.4 V	GND			GND	"	
	$V_{SD(LO)}$	57	"	"		"	"		"	"		0.4 V	"			"	"	
5/ T_C = +125°C	A_{VS}	58	2.5 V	2.5 V		"	"			"							"	
			"	"		"	"			"							"	
	$f_{OSC(MAX)}$	59	"	"		"	"	2 kΩ	0.001 µF	"							"	
	f_{OSC2}	60	"	"		"	"	"	3/	"							"	
	V_{OSC}	61	"	"		"	"	"	.01 µF	"							"	
	V_{RAMP}	62	"	"		"	"	"	"	"							"	
	V_{SEN}	63	2.425 V	2.575 V			"	"	"	"							"	
	$V_{SD(HI)}$	64	"	"		GND	"		"	"		1.4 V	GND			GND	"	
	$V_{SD(LO)}$	65	"	"		"	"		"	"		0.4 V	"			GND	"	
6/ T_C = -55°C		66 to 73	All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 5, T_C = +125°C.															

See footnotes at end of table.

TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
4 T_C = +25°C	A_{VS}	48	K1, K2, K9, K10, K11 / K1, K2, K9, K10, K11	9 / 9	E37 / E38	V / V	$A_{VS} = 20 \text{ Log } \left[\frac{\Delta V_{IN}\ (101)}{\lvert E37-E38 \rvert}\right]$ 2/	72		dB
	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$	49	K16	16	E39	V rms	$\frac{\Delta V_{IN}}{\Delta V_{REF}} = 20 \text{ Log } \left(\frac{1 \text{ V rms}}{E39 \text{ rms}}\right)$	50		dB
	$f_{OSC(MAX)}$	50	K4, K6	3	f1	kHz	$f_{OSC(MAX)} = f1$	250		kHz
	f_{OSC1}	51	"	"	f2	"	$f_{OSC1} = f2$	47	58	kHz
	$\frac{\Delta f_{OSC1}}{\Delta V_{IN}}$	52	" / "	" / "	f3 / f4	" / "	$\frac{\Delta f_{OSC1}}{\Delta V_{IN}} = 100 \quad \frac{(f3-f4)}{f3}$	-2	2	%
	V_{OSC}	53	"	"	E40	V	$V_{OSC} = E40$	2.4		V
	V_{RAMP}	54	"	7	E41	"	$V_{RAMP} = E41$	3.0	3.8	V
	V_{SEN}	55		4	E42	"	$V_{SEN} = E42$ 4/	190	210	mV
	$V_{SD(HI)}$	56	K6	9	E43	"	$V_{SD(HI)} \geq 1.4$ V if E43 $\leq$ limit		0.5	V
	$V_{SD(LO)}$	57	K6	"	E44	"	$V_{SD(LO)} \leq 0.4$ V if E44 $\geq$ limit	3.6		V
5 T_C = +125°C	A_{VS}	58	K1, K2, K9, K10, K11 / K1, K2, K9, K10, K11	" / "	E45 / E46	" / "	$A_{VS} = 20 \text{ Log } \left[\frac{\Delta V_{IN}\ (101)}{\lvert E45-E46 \rvert}\right]$ 2/	68		dB
	$f_{OSC(MAX)}$	59	K4, K6	3	f5	kHz	$f_{OSC(MAX)} = f5$	250		kHz
	f_{OSC2}	60	"	"	f6	kHz	$f_{OSC2} = f6$	45	60	kHz
	V_{OSC}	61	"	"	E47	V	$V_{OSC} = E47$	2.4		V
	V_{RAMP}	62	"	7	E48	V	$V_{RAMP} = E48$	3.0	3.8	V
	V_{SEN}	63		4	E49	V	$V_{SEN} = E49$ 4/	165	235	mV
	$V_{SD(HI)}$	64	K6	9	E50	"	$V_{SD(HI)} \geq 1.4$ V if E50 $\leq$ limit		0.5	V
	$V_{SD(LO)}$	65	K6	9	E51	"	$V_{SD(LO)} \leq 0.4$ V if E51 $\geq$ limit	3.6		V
6 T_C = -55°C		66 to 73								

See footnotes at end of table.

TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			INV INPUT	NONINV INPUT	OSC	+CL	-CL	R_T	C_T	GND	COMP	SHUT-DOWN	E_A	C_A	C_B	E_B	V_{IN}	V_{REF}
7 T_C = +25°C	G_{BW}	74																
	N_O	75	2.5 V	2.5 V		GND	GND			GND							20 V	
9 T_C = +25°C	t_{PW}	76	"	"		"	"	2 kΩ	0.01 µF	"			GND	20 V	20 V	GND	"	
	$t_{ON(A)}^{MAX}$ / f_{OSC}	77	V_{CM} = 2.5 V			"	"	"	"	"	3.8 V		"	"	"	"	"	
	$t_{ON(A)}^{MIN}$ / f_{OSC}	78	"			"	"	"	"	"	0.5 V		"	"	"	"	"	
	$t_{ON(B)}^{MAX}$ / f_{OSC}	79	"			"	"	"	"	"	3.8 V		"	"	"	"	"	
	$t_{ON(B)}^{MIN}$ / f_{OSC}	80	"			"	"	"	"	"	0.5 V		"	"	"	"	"	
	$t_{R(tr)A}$	81	2.425 V	2.575 V		"	"	"	"	"			"	"	"	"	"	
	$t_{R(tf)A}$	82	"	"		"	"	"	"	"			"	"	"	"	"	
	$t_{R(tr)B}$	83	"	"		"	"	"	"	"			"	"	"	"	"	
	$t_{R(tf)B}$	84	"	"		"	"	"	"	"			"	"	"	"	"	
10 T_C = +125°C	t_{PW}	85	2.5 V	2.5 V		"	"	"	"	"			"	"	"	"	"	
	$t_{ON(A)}^{MAX}$ / f_{OSC}	86	V_{CM} = 2.5 V			"	"	"	"	"	3.8 V		"	"	"	"	"	
	$t_{ON(A)}^{MIN}$ / f_{OSC}	87	"			"	"	"	"	"	0.5 V		"	"	"	"	"	
	$t_{ON(B)}^{MAX}$ / f_{OSC}	88	"			"	"	"	"	"	3.8 V		"	"	"	"	"	
	$t_{ON(B)}^{MIN}$ / f_{OSC}	89	"			"	"	"	"	"	0.5 V		"	"	"	"	"	
11 T_C = -55°C		90 to 94	(All test parameters, test conditions, equations and test limits are identical with those specified in table III, subgroup 10, T_C = +125°C).															

1/ V_{OSC} = 5 V, 50 µs pulse. Pulse V_{OSC} to switch output transistor (which is on).
2/ Measure V_{COMP} with V_N = 1.8 V and V_N = 3.4 V.
3/ The capacitor shall be polystyrene 0.01 µF tuned with a mica capacitor to ±0.1%.
4/ Adjust V +C.L. sense until V_{COMP} = 2 V.
5/ See figure 6.

TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
7 T_C = +25°C	G_{BW}	74					See figure 7	2		MHz
	N_O	75		16	E59	µV rms	N_O = E59		200	µV rms
9 T_C = +25°C	t_{PW}	76	K4, K6, K12, K13	3	t1	µs	t_{PW} = t1 5/	0.3		µs
	$\frac{t_{ON(A)}^{MAX}}{t_{OSC}}$	77	"	12	t2, t_{per}	"	$\frac{t_{ON(A)\ MAX}}{t_{OSC}} = \frac{t2}{t_{per}} \times 100$ 5/	45	50	%
	$\frac{t_{ON(A)}^{MIN}}{t_{OSC}}$	78	"	12	t3, t_{per}	"	$\frac{t_{ON(A)\ MIN}}{t_{OSC}} = \frac{t3}{t_{per}} \times 100$ 5/		0.001	%
	$\frac{t_{ON(B)}^{MAX}}{t_{OSC}}$	79	"	13	t4, t_{per}	"	$\frac{t_{ON(B)\ MAX}}{t_{OSC}} = \frac{t4}{t_{per}} \times 100$ 5/	45	50	%
	$\frac{t_{ON(B)}^{MIN}}{t_{OSC}}$	80	"	13	t5, t_{per}	"	$\frac{t_{ON(B)\ MIN}}{t_{OSC}} = \frac{t5}{t_{per}} \times 100$ 5/		0.001	%
	$t_{R(tr)A}$	81	"	12	t6	"	$t_{R(tr)A}$ = t6 5/		0.4	µs
	$t_{R(tf)A}$	82	"	12	t7	"	$t_{R(tf)A}$ = t7 5/		0.2	"
	$t_{R(tr)B}$	83	"	13	t8	"	$t_{R(tr)B}$ = t8 5/		0.4	"
	$t_{R(tf)B}$	84	"	13	t9	"	$t_{R(tf)B}$ = t9 5/		0.2	"
10 T_C = +125°C	t_{PW}	85	"	3	t10	"	t_{PW} = t10 5/	0.3		"
"	$\frac{t_{ON(A)}^{MAX}}{t_{OSC}}$	86	"		t11, t_{per}	"	$\frac{t_{ON(A)\ MAX}}{t_{OSC}} = \frac{t11}{t_{per}} \times 100$ 5/	45	50	%
	$\frac{t_{ON(A)}^{MIN}}{t_{OSC}}$	87	"		t12, t_{per}	"	$\frac{t_{ON(A)\ MIN}}{t_{OSC}} = \frac{t12}{t_{per}} \times 100$ 5/		0.001	%
	$\frac{t_{ON(B)}^{MAX}}{t_{OSC}}$	88	"		t13, t_{per}	"	$\frac{t_{ON(B)\ MAX}}{t_{OSC}} = \frac{t13}{t_{per}} \times 100$ 5/	45	50	%
	$\frac{t_{ON(B)}^{MIN}}{t_{OSC}}$	89	"		t14, t_{per}	"	$\frac{t_{ON(B)\ MIN}}{t_{OSC}} = \frac{t14}{t_{per}} \times 100$ 5/		0.001	%
11 T_C = -55°C		90 to 94								

TABLE III. Group A inspection for device types 02 and 04.

Subgroup	Symbol	Test no.	Adapter pin numbers															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			(-) INPUT	(+) INPUT	Sync	OSC	C_T	R_T	DIS-CHG	SS	COMP	SHUT-DOWN	O/P A	GND	V_C	O/P B	V_{IN}	V_{REF}
1 T_C = +25°C	V_{REF}	1	2.5 V	2.5 V										GND			20 V	
	V_{RLINE}	2	"	"										"			8 V	
			"	"										"			35 V	
	V_{RLOAD}	3	"	"										"			20 V	-20 mA
	I_{OS}	4	"	"										"			20 V	0 V
	I_C	5	"	"								1.6 V		"	35 V		30 V	
	$V_{OL(A)}$	6	GND	V_{REF}		2/				-			20 mA	"	20 V		20 V	
	$V_{OL(A)}$	7	"	"		"							100 mA	"	"		"	
	$V_{OL(B)}$	8	"	"		"								"	"	20 mA	"	
	$V_{OL(B)}$	9	"	"		"								"	"	100 mA	"	
	$V_{OH(A)}$	10	"	"		"							-20 mA	"	"		"	
	$V_{OH(A)}$	11	"	"		"							-100 mA	"	"		"	
	$V_{OH(B)}$	12	"	"		"								"	"	-20 mA	"	
	$V_{OH(B)}$	13	"	"		"								"	"	-100 mA	"	
	V_{IO}	14	2.5 V	2.5 V										"			"	
	$+I_{IB}$	15	"	"										"			"	
	$-I_{IB}$	16	"	"										"			"	
	I_{IO}	17																
	C_{MR}	18	V_{CM}	V_{CM}										GND			20 V	
	I_{IN}	19	2.5 V	2.5 V										"			35 V	
	$V_{SAT(OSC)}$	20	2.5 V	2.5 V		5 V			15 mA					"			20 V	
	V_{HI}	21	GND	5.1 V										"			"	
	V_{LO}	22	5.1 V	GND										"			"	

See footnotes at end of table.

TABLE ... Group A inspection for device type 01 - Continued

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
1 T_C = +25°C	V_{REF}	1		16	E1	V	V_{REF} = E1	5.05	5.15	V
	V_{RLINE}	2		"	E2	"	V_{RLINE} = E3-E2	-30	30	mV
				"	E3	"				
	V_{RLOAD}	3		"	E4	"	V_{RLOAD} = E4-E1	-50	50	mV
	I_{OS}	4		"	I1	mA	I_{OS} = I1	-125		mA
	I_C	5		13	I2	µA	I_C = I2 1/		200	µA
	$V_{OL(A)}$	6		11	E5	V	$V_{OL(A)}$ = E5		0.4	V
	$V_{OL(A)}$	7		11	E6	"	$V_{OL(A)}$ = E6		2.0	"
	$V_{OL(B)}$	8		14	E7	"	$V_{OL(B)}$ = E7		0.4	"
	$V_{OL(B)}$	9		14	E8	"	$V_{OL(B)}$ = E8		2.0	"
	$V_{OH(A)}$	10		11	E9	"	$V_{OH(A)}$ = E9	18		"
	$V_{OH(A)}$	11		11	E10	"	$V_{OH(A)}$ = E10	17		"
	$V_{OH(B)}$	12		14	E11	"	$V_{OH(B)}$ = E11	18		"
	$V_{OH(B)}$	13		14	E12	"	$V_{OH(B)}$ = E12	17		"
	V_{IO}	14	K12 (pos. 3), K13, K15	9	E13	"	$V_{IO} = \frac{E13-2.5}{101}$	-5	5	mV
	$+I_{IB}$	15	K3, K12 (pos. 3), K13, K15	"	E14	"	$+I_{IB} = \frac{E13-E14}{101000}$	0.01	10	µA
	$-I_{IB}$	16	K2, K12 (pos. 3), K13, K15	"	E15	"	$-I_{IB} = \frac{E15-E13}{101000}$	0.01	10	µA
	I_{IO}	17					$I_{IO} = (+I_{IB}) - (-I_{IB})$	-1	1	µA
	C_{MR}	18	K12 (pos. 3), K13, K15	9	E16	V	3/	60		dB
	I_{IN}	19		15	I3	mA	I_{IN} = I3		20	mA
	$V_{SAT(OSC)}$	20		5	E17	V	$V_{SAT(OSC)}$ = E17	0.5	1.1	V
	V_{HI}	21	K12 (pos. 2)	9	E18	"	V_{HI} = E18	3.8		"
	V_{LO}	22	"	"	E19	"	V_{LO} = E19		0.5	"

See footnotes at end of table.

TABLE III. Group A inspection for device types 02 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			(-) INPUT	(+) INPUT	Sync	OSC	C_T	R_T	DIS-CHG	SS	COMP	SHUT-DOWN	O/P A	GND	V_C	O/P B	V_{IN}	V_{REF}
2 T_C = +125°C	V_{REF}	23	2.5 V	2.5 V										GND			20 V	
	V_{RLINE}	24	"	"										"			8 V	
			"	"										"			35 V	
	V_{RLOAD}	25	"	"										"			20 V	-20 mA
	I_{OS}	26	"	"										"			20 V	0 V
	I_C	27	"	"								1.6 V		"	35 V		30 V	
	$V_{OL(A)}$	28	GND	V_{REF}		2/							20 mA	"	20 V		20 V	
	$V_{OL(A)}$	29	"	"		"							100 mA	"	"		"	
	$V_{OL(B)}$	30	"	"		"								"	"	20 mA	"	
	$V_{OL(B)}$	31	"	"		"								"	"	100 mA	"	
	$V_{OH(A)}$	32	"	"		"							-20 mA	"	"		"	
	$V_{OH(A)}$	33	"	"		"							-100 mA	"	"		"	
	$V_{OH(B)}$	34	"	"		"								"	"	-20 mA	"	
	$V_{OH(B)}$	35	"	"		"								"	"	-100 mA	"	
	V_{IO}	36	2.5 V	2.5 V										"			"	
	$+I_{IB}$	37	"	"										"			"	
	$-I_{IB}$	38	"	"										"			"	
	I_{IO}	39																
	C_{MR}	40	V_{CH}	V_{CH}										GND			20 V	
	I_{IH}	41	2.5 V	2.5 V										"			35 V	
	V_{HI}	42	GND	5.1 V										"			20 V	
	V_{LO}	43	5.1 V	GND										"			20 V	
3 T_C = -55°C		44 to 64	All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 2, T_C = +125°C. Except for I_{IH} at -55°C. The maximum limit is 20 mA.)															

See footnotes at end of table.

TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
2 TC = +125°C	V_{REF}	23		16	E20	V	V_{REF} = E20	5.0	5.2	V
	V_{RLINE}	24		"	E21	"	V_{RLINE} = E22-E21	-30	30	mV
				"	E22	"				
	V_{RLOAD}	25		"	E23	"	V_{RLOAD} = E23-E20	-50	50	mV
	I_{OS}	26		"	I4	mA	I_{OS} = I4	-125		mA
	I_C	27		13	I5	µA	I_C = I5 1/		200	µA
	$V_{OL(A)}$	28		11	E24	V	$V_{OL(A)}$ = E24		0.4	V
	$V_{OL(A)}$	29		11	E25	"	$V_{OL(A)}$ = E25		2.0	"
	$V_{OL(B)}$	30		14	E26	"	$V_{OL(B)}$ = E26		0.4	"
	$V_{OL(B)}$	31		14	E27	"	$V_{OL(B)}$ = E27		2.0	"
	$V_{OH(A)}$	32		11	E28	"	$V_{OH(A)}$ = E28	18		"
	$V_{OH(A)}$	33		11	E29	"	$V_{OH(A)}$ = E29	17		"
	$V_{OH(B)}$	34		14	E30	"	$V_{OH(B)}$ = E30	18		"
	$V_{OH(B)}$	35		14	E31	"	$V_{OH(B)}$ = E31	17		"
	V_{IO}	36	K12 (pos. 3), K13, K15	9	E32	"	$V_{IO} = \frac{E32-2.5}{101}$	-5	5	mV
	$+I_{IB}$	37	K3, K12 (pos. 3), K13, K15	"	E33	"	$+I_{IB} = \frac{E32-E33}{101000}$	0.01	10	µA
	$-I_{IB}$	38	K2, K12 (pos. 3), K13, K15	"	E34	"	$-I_{IB} = \frac{E34-E32}{101000}$	0.01	10	µA
	I_{IO}	39					$I_{IO} = (+I_{IB}) - (-I_{IB})$	-1	1	µA
	C_{MR}	40	K12 (pos. 3), K13, K15	9	E35	V	3/	60		dB
	I_{IN}	41		15	I6	mA	I_{IN} = I6		18	mA
	V_{HI}	42	K12 (pos. 2)	9	E36	"	V_{HI} = E36	3.8		V
	V_{LO}	43	K12 (pos. 2)	9	E37	"	V_{LO} = E37		0.5	V
3 T_C = -55°C		44 to 64								

See footnotes at end of table.

TABLE III. Group A inspection for device types 02 and 3e - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			(-) INPUT	(+) INPUT	Sync	OSC	C_T	R_T	DIS-CHG	SS	COMP	SHUT-DOWN	O/P A	GND	V_C	O/P B	V_{IN}	V_{REF}
4 T_C = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$	65	2.5 V	2.5 V										GND			20 V +1 V rms	
	f_{OSC}	66	"	"			0.01 µF	3.6 kΩ	0 Ω					"			20 V	0.01 µF to GND
	V_{OSC}	67	"	"			"	"	"					"			"	0.01 µF to GND
	V_{RAMP}	68	"	"			"	"	"					"			"	"
	$\frac{\Delta f_{OSC1}}{\Delta V_{IN}}$	69	"	"			"	"	"					"			8 V	"
			"	"			"	"	"					"			35 V	"
	$f_{OSC(MIN)}$	70	"	"			0.1 µF	150 kΩ	"					"			20 V	"
	$f_{OSC(MAX)}$	71	"	"			0.001 µF	2 kΩ	"					"			"	"
	$V_{SYNC(LO)}$	72	"	"										"			"	"
	$V_{SYNC(HI)}$	73	"	"										"			"	"
	I_{SYNC}	74	"	"	3.5 V									"			"	"
	I_{SS}	75	"	"						0 V		0 V		"			"	
	I_{SD}	76	"	"								2.5 V		"			"	
	V_{SS}	77	"	"								2 V		"			"	
	$V_{SD(LO)}$	78	"	"			0.01 µF	3.6 kΩ	0 Ω		3.6 V	6/		"	20 V		"	
	$V_{SD(HI)}$	79	"	"			0.01 µF	3.6 kΩ	0 Ω		3.6 V	7/		"	20 V		"	
	A_{VS}	80	"	"										"			"	
	P_{SRR}	81	5.1 V	5.1 V										"			8 V	
														"			35 V	
5 T_C = +125°C	f_{OSC2}	82	2.5 V	2.5 V			0.01 µF	3.6 kΩ	0 Ω					"			20 V	0.01 µF to GND
	V_{OSC}	83	"	"			"	"	"					"			"	"
	V_{RAMP}	84	"	"			"	"	"					"			"	"
	$f_{OSC(MIN)}$	85	"	"			0.1 µF	150 kΩ	"					"			"	"
	$f_{OSC(MAX)}$	86	"	"			0.001 µF	2 kΩ						"			"	"

See footnotes at end of table.

TABLE III. Group A inspection for device types 02 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers																
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
			(-) INPUT	(+) INPUT	SYNC	OSC	C_T	R_T	DIS-CHG	SS	COMP	SHUT-DOWN	O/P A	GND	V_C	O/P B	V_{IN}	V_{REF}	
T_C = +125°C 5	$V_{SYNC(LO)}$	87	2.5 V	2.5 V										GND			20 V	0.01 µF to GND	
	$V_{SYNC(HI)}$	88	"	"										"			"	"	
	I_{SYNC}	89	"	"	3.5 V									"			"	"	
	I_{SS}	90	"	"						0 V		0 V		"			"		
	I_{SD}	91	"	"								2.5 V		"			"		
	V_{SS}	92	"	"								2 V		"			"		
	$V_{SD(LO)}$	93	"	"			0.01 µF	3.6 kΩ	0 Ω		3.6 V	6/		"	20 V		"		
	$V_{SD(HI)}$	94	"	"			0.01 µF	3.6 kΩ	0 Ω		3.6 V	7/		"	20 V		"		
	A_{VS}	95	"	"										"			"		
	P_{SRR}	96	5.1 V	5.1 V										"			8 V		
														"			35 V		
T_C = -55°C 6		97 to 112	All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 5, T_C = +125°C.																
T_C = +25°C 7	G_{BW}	113	θ_1											GND			20 V		
	N_O	114	2.5 V	2.5 V										"			"		
T_C = +25°C 9	t_{PW}	115	"	"			0.01 µF	3.6 kΩ	0 Ω					"			"		
	$t_{ON(A)\ MAX}$ / V_{OSC}	116	"	"			"	"	"		3.6 V	GND	4/	"	20 V	4/	"		
	$t_{ON(B)\ MAX}$ / V_{OSC}	117	"	"			"	"	"		3.6 V	"	"	"	"	"	"		
	$t_{ON(A)\ MIN}$ / V_{OSC}	118	"	"			"	"	"		0.6 V	"	"	"	"	"	"		
	$t_{ON(B)\ MIN}$ / V_{OSC}	119	"	"			"	"	"		0.6 V	"	"	"	"	"	"		
	V_{UL}	120	GND	V_{REF}		2/								"	V_{IN}		9/		
	$t_{SD(A)}$	121	"	"		"								"	20 V		20 V		
	$t_{SD(B)}$	122	"	"		"								"	20 V		20 V		

See footnotes at end of table.

TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	Test no.	Adapter pin numbers																	
			1 (+) ERROR	2 (-) ERROR	3 COMP	4 C_{SS}	5 RESET	6 -C.S.	7 +C.S.	8 SHUT-DOWN	9 R_T	10 C_T	11 R_D	12 SYNC	13 O/P A	14 V_C	15 GND	16 O/P B	17 V_{IN}	18 V_{REF}
1 T_C = +25°C	V_{REF}	1	GND	GND				GND	GND								GND		15 V	
	V_{RLINE}	2	"	"				"	"								"		8 V	
			"	"				"	"								"		35 V	
	V_{RLOAD}	3	"	"				"	"								"		15 V	-20 mA
	I_{OS}	4	"	"				"	"								"		"	0 V
	$V_{CE(SAT)A}$	5	"	"				"	"	GND					20 mA	15 V	"		"	
		6	"	"				"	"	"					100 mA	"	"		"	
	$V_{CE(SAT)B}$	7	"	"				"	"	"						"	"	20 mA	"	
		8	"	"				"	"	"						"	"	100 mA	"	
	$V_{OH(A)}$	9	V_{REF}	"				"	"			11/			-20 mA	"	"		"	
		10	"	"				"	"			"			-100 mA	"	"		"	
	$V_{OH(B)}$	11	"	"				"	"			"				"	"	-20 mA	"	
		12	"	"				"	"			"				"	"	-100 mA	"	
	I_C	13	GND	"						GND						35 V	"		"	
	I_{IN}	14	GND	"						0.4 V							"		35 V	
	V_{IO}	15	2.5 V R1 = 100Ω	2.5 V R2 = 100Ω													"		15 V	
	$+I_{IB}$	16	"	"													"		"	
	$-I_{IB}$	17	"	"													"		"	

See footnotes at end of table.

TABLE III. Group A inspection for device types 02 and 04 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
9 T_C = +25°C	$t_{R(tr)A}$	123	K4, K5, K6, K7, K11, K12, (pos. 2)	11	t8	ns	$t_{R(tr)A}$ = t8 see figure 6		600	ns
	$t_{R(tf)A}$	124	"	11	t9	"	$t_{R(tf)A}$ = t9 see figure 6		300	"
	$t_{R(tr)B}$	125	"	14	t10	"	$t_{R(tr)B}$ = t10 see figure 6		600	"
	$t_{R(tf)B}$	126	"	14	t11	"	$t_{R(tf)B}$ = t11 see figure 6		300	"
10 T_C = +125°C	t_{PW}	127	K4, K5, K6, K7	4	t12	µs	t_{PW} = t12 see figure 6	0.3	1.4	µs
	$\frac{t_{ON(A)}^{MAX}}{t_{OSC}}$	128	K4, K5, K6, K7, K11, K12, (pos. 2)	11	t13, t_{per}	"	$\frac{t_{ON(A)}\ MAX}{t_{OSC}} = \frac{t13}{t_{per}} \times 100$ see figure 6	45	50	%
	$\frac{t_{ON(B)}^{MAX}}{t_{OSC}}$	129	"	14	t14, t_{per}	"	$\frac{t_{ON(B)}\ MAX}{t_{OSC}} = \frac{t14}{t_{per}} \times 100$ see figure 6	45	50	"
	$\frac{t_{ON(A)}^{MIN}}{t_{OSC}}$	130	"	11	t15, t_{per}	"	$\frac{t_{ON(A)}\ MIN}{t_{OSC}} = \frac{t15}{t_{per}} \times 100$ see figure 6		0.001	"
	$\frac{t_{ON(B)}^{MIN}}{t_{OSC}}$	131	"	14	t16, t_{per}	"	$\frac{t_{ON(B)}\ MIN}{t_{OSC}} = \frac{t16}{t_{per}} \times 100$ see figure 6		0.001	"
	V_{UL}	132		15	E77	V	V_{UL} = E77	6	8	V
	$t_{SD(A)}$	133		11	t17	ns	See figure 6 10/		700	ns
	$t_{SD(B)}$	134		14	t18	"	See figure 6 10/		700	"
	$t_{R(tr)A}$	135	K4, K5, K6, K7, K11, K12, (pos. 2)	11	t19	"	$t_{R(tr)A}$ = t19 see figure 6		600	ns
	$t_{R(tf)A}$	136	"	11	t20	"	$t_{R(tf)A}$ = t20 see figure 6		300	"
	$t_{R(tr)B}$	137	"	14	t21	"	$t_{R(tr)B}$ = t21 see figure 6		600	"
	$t_{R(tf)B}$	138	"	14	t22	"	$t_{R(tf)B}$ = t22 see figure 6		300	"
11 T_C = -55°C		139 to 151								

See footnotes at end of table.

TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
1 T_C = +25°C	V_{REF}	1		18	E1	V	V_{REF} = E1	4.9	5.1	V
	V_{RLINE}	2		"	E2	"	V_{RLINE} = E3-E2	-20	20	mV
				"	E3	"				
	V_{RLOAD}	3		"	E4	"	V_{RLOAD} = E4-E1	-30	30	mV
	I_{OS}	4		"	I1	mA	I_{OS} = I1	-125		mA
	$V_{CE(SAT)A}$	5		13	E5	V	$V_{CE(SAT)A}$ = E5		0.3	V
		6		13	E6	"	$V_{CE(SAT)A}$ = E6		2.0	"
	$V_{CE(SAT)B}$	7		16	E7	"	$V_{CE(SAT)B}$ = E7		0.3	"
		8		16	E8	"	$V_{CE(SAT)B}$ = E8		2.0	"
	$V_{OH(A)}$	9		13	E9	"	$V_{OH(A)}$ = E9	12.5		"
		10		13	E10	"	$V_{OH(A)}$ = E10	12.0		"
	$V_{OH(B)}$	11		16	E11	"	$V_{OH(B)}$ = E11	12.5		"
		12		16	E12	"	$V_{OH(B)}$ = E12	12.0		"
	I_C	13		14	I2	µA	I_C = I2		150	µA
	I_{IN}	14		17	I3	mA	I_{IN} = I3		30	mA
	V_{IO}	15	K3 (pos. 3), K4, K6	3	E13	V	V_N = 2.5 V $V_{IO} = \frac{(E13-2.5\ V)}{201}$	-5	5	mV
	$+I_{IB}$	16	K1, K3 (pos. 3), K4, K6	"	E14	"	V_N = 2.5 V $+I_{IB} = \frac{E13-E14}{201000}$	-1		µA
	$-I_{IB}$	17	K2, K3 (pos. 3), K4, K6	"	E15	"	V_N = 2.5 V $-I_{IB} = \frac{E15-E13}{201000}$	-1		µA

See footnotes at end of table.

TABLE III. Group A inspection for device types 02 and 04 - Continued.

Subgroup	Symbol	Test No.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
5 T_C = +125°C	$V_{SYNC(LO)}$	87					See figure 6 5/		0.8	V
	$V_{SYNC(HI)}$	88					See figure 6 5/	2.8		V
	I_{SYNC}	89		3	I13	mA	I_{SYNC} = I13		2.5	mA
	I_{SS}	90		8	I14	µA	I_{SS} = I14	-80	-25	µA
	I_{SD}	91		10	I15	mA	I_{SD} = I15		1	mA
	V_{SS}	92		8	E65	V	V_{SS} = E65		0.6	V
	$V_{SD(LO)}$	93					6/		0.5	"
	$V_{SD(HI)}$	94					7/	1.6		"
	A_{VS}	95	K12 (pos. 3), K13, K15	9	E66	V	V_N = 1 V to 3.5 V A_{VS} = 20 Log $\left[\frac{\Delta V_N(10)}{\Delta E66}\right]$	60		dB
	P_{SRR}	96	"	"	E67	"	8/	60		dB
			"	"	E68	"				
6 T_C = -55°C		97 to 112								
7 T_C = +25°C	G_{BW}	113					G_{BW} = f_{e1}, when e_0 = 20 mV see figure 7	2		MHz
	N_O	114		16	E75	µV rms	N_O = E75		200	µV rms
9 T_C = +25°C	t_{PW}	115	K4, K5, K6, K7	4	t1	µs	t_{PW} = t1 see figure 6	0.3	1.0	µs
	$\frac{t_{ON(A)}}{t_{OSC}}$ MAX	116	K4, K5, K6, K7, K11, K12 (pos. 2)	11	t2, t_{per}	"	$\frac{t_{ON(A)}}{t_{OSC}}$ MAX = $\frac{t2}{t_{per}}$ X 100 see figure 6	45	50	%
	$\frac{t_{ON(B)}}{t_{OSC}}$ MAX	117	"	14	t3, t_{per}	"	$\frac{t_{ON(B)}}{t_{OSC}}$ MAX = $\frac{t3}{t_{per}}$ X 100 see figure 6	45	50	"
	$\frac{t_{ON(A)}}{t_{OSC}}$ MIN	118	"	11	t4, t_{per}	"	$\frac{t_{ON(A)}}{t_{OSC}}$ MIN = $\frac{t4}{t_{per}}$ X 100 see figure 6		0.001	"
	$\frac{t_{ON(B)}}{t_{OSC}}$ MIN	119	"	14	t5, t_{per}	"	$\frac{t_{ON(B)}}{t_{OSC}}$ MIN = $\frac{t5}{t_{per}}$ X 100 see figure 6		0.001	"
	V_{UL}	120		15	E76	V	V_{UL} = E76	6	6	V
	$t_{SD(A)}$	121		11	t6	ns	See figure 6 10/		500	ns
	$t_{SD(B)}$	122		14	t7	ns	See figure 6 10/		500	ns

See footnotes at end of table.

TABLE III. Group A inspection for device types 02 and 04 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
4 T_C = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$	65	K1	16	E56	μV rms	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$ = 20 Log $\left(\frac{\Delta V_{IN}}{\Delta E56}\right)$	50		dB
	f_{OSC}	66	K4, K5, K6, K7	4	f1	kHz	f_{OSC} = f1	37.5	42.5	kHz
	V_{OSC}	67	"	4	E57	V	V_{OSC} = E57	3.0		V
	V_{RAMP}	68	"	5	E58	V	V_{RAMP} = E58	3.0	3.6	V
	$\frac{\Delta f_{OSC1}}{\Delta V_{IN}}$	69	"	4	f2	kHz	$\frac{\Delta f_{OSC1}}{\Delta V_{IN}}$ = 100 $\left(\frac{(f2-f3)}{Y2}\right)$	-1	1	%
			"	"	f3	kHz				
	$f_{OSC(MIN)}$	70	"	"	f4	Hz	$f_{OSC(MIN)}$ = f4		150	Hz
	$f_{OSC(MAX)}$	71	"	"	f5	kHz	$f_{OSC(MAX)}$ = f5	300		kHz
	$V_{SYNC(LO)}$	72					See figure 6 5/		0.8	V
	$V_{SYNC(HI)}$	73					See figure 6 5/	2.8		V
	I_{SYNC}	74		3	I10	mA	I_{SYNC} = I10		2.5	mA
	I_{SS}	75		8	I11	μA	I_{SS} = I11	-80	-25	μA
	I_{SD}	76		10	I12	mA	I_{SD} = I12		1	mA
	V_{SS}	77		8	E59	V	V_{SS} = E59		0.6	V
	$V_{SD(LO)}$	78	K4, K5, K6, K7	11 and 14			6/		0.5	"
	$V_{SD(HI)}$	79	K4, K5, K6, K7	11 and 14			7/	1.6		"
	A_{VS}	80	K12 (pos. 3), K13, K15	9	E60	V	V_N = 1 V to 3.5 V A_{VS} = 20 Log $\left[\frac{\Delta V_N(101)}{\Delta E60}\right]$	60		dB
	P_{SRR}	81	"	"	E61	"	8/	60		dB
			"	"	E62	"				
5 T_C = +125°C	f_{OSC2}	82	K4, K5, K6, K7	4	f6	kHz	f_{OSC2} = f6	36	44	kHz
	V_{OSC}	83	"	4	E63	V	V_{OSC} = E63	3.0		V
	V_{RAMP}	84	"	5	E64	V	V_{RAMP} = E64	3.0	3.6	V
	$f_{OSC(MIN)}$	85	"	4	f7	Hz	$f_{OSC(MIN)}$ = f7		150	Hz
	$f_{OSC(MAX)}$	86	"	4	f8	kHz	$f_{OSC(MAX)}$ = f8	300		kHz

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers																	
			1 (+) ERROR	2 (-) ERROR	3 COMP	4 C_{SS}	5 RESET	6 -C.S.	7 +C.S.	8 SHUT-DOWN	9 R_T	10 C_T	11 R_D	12 SYNC	13 O/P A	14 V_C	15 GND	16 O/P B	17 V_{IN}	18 V_{REF}
T_C = +25°C	I_{IO}	18															GND			
	C_{MR}	19	0 V 100Ω	0 V 100Ω													"		15 V	
			5.2 V 100Ω	5.2 V 100Ω													"		"	
	$I_{IH(\overline{SD})}$	20	GND	GND				GND	GND	2.4 V to ammeter							"		"	
	$I_{IL(\overline{SD})}$	21	"	"				"	"	0.4 V to ammeter							"		"	
	$V_{OHP(\overline{SD})}$	22	"	"				"	"	-40 µA							"		"	
	$V_{OLP(\overline{SD})}$	23	"	"				"	120 mV	3.6 mA							"		"	
	$I_{IH(\overline{R})}$	24	"	"			2.4 V to ammeter	"	GND								"		"	
	$I_{IL(\overline{R})}$	25	"	"			0.4 V to ammeter	"	"								"		"	
	$V_{OHP(\overline{R})}$	26	"	"			-40 µA	"	"								"		"	
	$V_{OLP(\overline{R})}$	27	"	"			3.6 mA	"	"								"		3.8 V	3.8 V
	$I_{IH(\overline{S})}$	28	"	"				"	"			≤ 0.5 V		2.4 V to ammeter			"		15 V	
	$I_{IL(\overline{S})}$	29	"	"				"	"			"		0.4 V to ammeter			"		"	
	$V_{OHP(\overline{S})}$	30	"	"				"	"			"		-40 µA			"		"	

See footnotes at end of table.

TABLE III. Group A inspection for device types 02 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
			(-) INPUT	(+) INPUT	Sync	OSC	C_T	R_T	DIS-CHG	SS	COMP	SHUT-DOWN	O/P A	GND	V_C	O/P B	V_{IN}	V_{REF}
9 T_C = +25°C	$t_{R(tr)A}$	123	2.5 V	2.5 V			0.01 µF	3.6 kΩ	0 Ω		3.6 V	GND		GND	20 V		20 V	
	$t_{R(tf)A}$	124	"	"			"	"	"		"	"		"	"		"	
	$t_{R(tr)B}$	125	"	"			"	"	"		"	"		"	"		"	
	$t_{R(tf)B}$	126	"	"			"	"	"		"	"		"	"		"	
10 T_C = +125°C	t_{PW}	127	"	"			"	"	"					"			"	
	$\frac{t_{ON(A)}^{MAX}}{t_{OSC}}$	128	"	"			"	"	"		3.6 V	GND	4/	"	20 V	4/	"	
	$\frac{t_{ON(B)}^{MAX}}{t_{OSC}}$	129	"	"			"	"	"		3.6 V	"	"	"	"	"	"	
	$\frac{t_{ON(A)}^{MIN}}{t_{OSC}}$	130	"	"			"	"	"		0.6 V	"	"	"	"	"	"	
	$\frac{t_{ON(B)}^{MIN}}{t_{OSC}}$	131	"	"			"	"	"		0.6 V	"	"	"	"	"	"	
	V_{UL}	132	GND	V_{REF}		2/								"	V_{IN}		9/	
	$t_{SD(A)}$	133	"	"		"								"	20 V		20 V	
	$t_{SD(B)}$	134	"	"		"								"	"		"	
	$t_{R(tr)A}$	135					0.01 µF	3.6 kΩ	0 Ω		3.6 V	GND		"	"		"	
	$t_{R(tf)A}$	136					"	"	"		3.6 V	"		"	"		"	
	$t_{R(tr)B}$	137					"	"	"		0.6 V	"		"	"		"	
	$t_{R(tf)B}$	138					"	"	"		0.6 V	"		"	"		"	
11 T_C = -55°C		139 to 151	All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 10, T_C = +125°C. Except for t_{PW} at -55°C, the maximum limit is 1.0 µs and for $t_{SD(A)}$ and $t_{SD(B)}$, the maximum limits are 500 ns.															

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
1 T_C = +25°C	I_{IO}	18					$I_{IO} = (-I_{IB}) - (+I_{IB})$	-0.5	0.5	µA
	C_{MR}	19	K3 (pos. 3), K4, K6	3	E16	V	12/	60		dB
			K3 (pos. 3), K4, K6	3	E17	"				
	$I_{IH(\overline{SD})}$	20		8	I4	µA	$I_{IH(\overline{SD})} = I4$	-200		µA
	$I_{IL(\overline{SD})}$	21		"	I5	µA	$I_{IL(\overline{SD})} = I5$	-360		µA
	$V_{OHP(\overline{SD})}$	22		"	E18	V	$V_{OHP(\overline{SD})} = E18$	2.4		V
	$V_{OLP(\overline{SD})}$	23		"	E19	V	$V_{OLP(\overline{SD})} = E19$		0.4	V
	$I_{IH(\overline{R})}$	24		5	I6	µA	$I_{IH(\overline{R})} = I6$	-200		µA
	$I_{IL(\overline{R})}$	25		"	I7	µA	$I_{IL(\overline{R})} = I7$	-360		µA
	$V_{OHP(\overline{R})}$	26		"	E20	V	$V_{OHP(\overline{R})} = E20$	2.4		V
	$V_{OLP(\overline{R})}$	27		"	E21	V	$V_{OLP(\overline{R})} = E21$		0.4	V
	$I_{IH(\overline{S})}$	28		12	I8	µA	$I_{IH(\overline{S})} = I8$	-200		µA
	$I_{IL(\overline{S})}$	29		"	I9	µA	$I_{IL(\overline{S})} = I9$	-360		µA
	$V_{OHP(\overline{S})}$	30		"	E22	V	$V_{OHP(\overline{S})} = E22$	2.4		V

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers																	
			1 (+) ERROR	2 (-) ERROR	3 COMP	4 C_{SS}	5 RESET	6 -C.S.	7 +C.S.	8 SHUT DOWN	9 R_T	10 C_T	11 R_D	12 SYNC	13 O/P A	14 V_C	15 GND	16 O/P B	17 V_{IN}	18 V_{REF}
1 T_C = +25°C	$V_{OLP(\overline{S})}$	31	GND	GND				GND	GND			> 3.6 V		3.6 mA			GND		15 V	
2 T_C = +125°C		32 to 62	All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 1, T_C = +25°C. Except for I_{OS}, the minimum limit is -100 mA and I_{IN} the maximum limit is 25 mA.																	
3 T_C = -55°C		63 to 93	All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 1, T_C = +25°C.																	
4 T_C = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$	94	GND	GND				GND	GND								GND		15 V +1 V rms at 2.4 kHz	
	f_{OSC}	95	"	"				"	"		4.12 kΩ	0.01 µF	0 Ω	2.7 kΩ to V_{REF}		15 V	"		15 V	
	V_{RV}	96	"	"				"	"		"	"	"	"			"		6 V	
	V_{RP}	97	"	"				"	"		"	"	"	"			"		35 V	
	$\frac{\Delta f_{OSC}}{\Delta V_{IN}}$	98	"	"				"	"		"	"	"	"		15 V	"		8 V	
			"	"				"	"		"	"	"	"		"	"		35 V	
	$f_{OSC(MIN)}$	99	"	"				"	"		150 kΩ	0.2 µF	"	"		"	"		15 V	
	$f_{OSC(MAX)}$	100	"	"				"	"		2 kΩ	1.0 nF	"	"		"	"		"	
	A_{VS}	101	2.5 V R1 = 100Ω	2.5 V R2 = 100Ω													"		"	
	P_{SRR}	102	"	"													"		8 V	
			"	"													"		35 V	
	V_S	103	150 mV	GND				GND									"		15 V	

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
1 T_C = +25°C	$V_{OLP(\overline{S})}$	31		12	E23	V	$V_{OLP(\overline{S})}$ = E23		0.4	V
2 T_C = +125°C		32 to 62								
3 T_C = -55°C		63 to 93								
4 T_C = +25°C	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$	94	K13	18	E70	μV rms	$\frac{\Delta V_{IN}}{\Delta V_{REF}}$ = 20 Log $\left(\frac{\Delta V_{IN}}{\Delta V_{REF}}\right)$	50		dB
	f_{OSC}	95	K5, K7, K9	13	f1	kHz	f_{OSC} = 2(f1)	38	42	kHz
	V_{RV}	96	"	10	E71	V	V_{RV} = E71	0.45		V
	V_{RP}	97	"	10	E72	V	V_{RP} = E72	2.5	3.5	V
	$\frac{\Delta f_{OSC}}{\Delta V_{IN}}$	98	"	13	f2	kHz	$\frac{\Delta f_{OSC}}{\Delta V_{IN}} = \frac{(f2-f3)}{f2} \times 100$	-1	1	%
			"	"	f3	kHz				
	$f_{OSC(MIN)}$	99	"	"	f4	Hz	$f_{OSC(MIN)} = [f4] \times 2$		100	Hz
	$f_{OSC(MAX)}$	100	"	16	f5	kHz	$f_{OSC(MAX)} = [f5] \times 2$	350		kHz
	A_{VS}	101	K3 (pos. 3), K4, K6	3	E73	V	13/	60		dB
	P_{SRR}	102	"	"	E74	"	14/	66		dB
			"	"	E75	"				
	V_S	103	K11, K12	7	E76	"	V_S = E76 15/	60	120	mV

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers																	
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
			(+) ERROR	(-) ERROR	COMP	C_{SS}	RESET	-C.S.	+C.S.	SHUT-DOWN	R_T	C_T	R_D	SYNC	O/P A	V_C	GND	O/P B	V_{IN}	V_{REF}
4 T_C = +25°C	$+I_{IBS}$	104	150 mV	GND				GND	GND								GND		15 V	
	$-I_{IBS}$	105	150 mV	"				"	"								"		"	
	V_{EC}	106	GND	"			0.4 V	"	"								"		"	
	I_{CS}	107	"	150 mV		0 V		"	"								"		"	
	$V_{R(LOW)}$	108	"	GND				"	"								"		3.8 V	3.8 V
	$V_{R(HIGH)}$	109	"	"				"	"								"		4.8 V	4.8 V
	V_{HI}	110	150 mV	"	-100 µA												"		15 V	
	V_{LO}	111	GND	150 mV	100 µA												"		"	
5 T_C = +125°C	f_{OSC2}	112	GND	GND				GND	GND		4.12 kΩ	0.01 µF	0 Ω	2.7 kΩ to V_{REF}			"		"	
		113 to 128	All test parameters, test conditions, equations, and test limits remaining are identical with tests 96 through 111 specified in table III, subgroup 4, T_C = +25°C.																	
6 T_C = -55°C		129 to 145	All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 5, T_C = +125°C.																	
7 T_C = +25°C	M_O	146	GND	GND				GND	GND								GND		15 V	
	G_{BW}	147	GND	GND													GND		15 V	

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
4 T_C = +25°C	$+I_{IBS}$	104	K11, K12	7	I28	µA	$+I_{IBS}$ = I28	-10		µA
	$-I_{IBS}$	105	K11, K12	6	I29	µA	$-I_{IBS}$ = I29	-10		µA
	V_{EC}	106		4	E77	V	V_{EC} = E77		0.4	V
	I_{CS}	107		4	I30	µA	I_{CS} = I30	-150	-50	µA
	$V_{R(LOW)}$	108		5	E78	V	$V_{R(LOW)}$ = E78		0.4	V
	$V_{R(HIGH)}$	109		5	E79	"	$V_{R(HIGH)}$ = E79	2.4		"
	V_{HI}	110	K3 (pos. 2)	3	E80	"	V_{HI} = E80	3.6		"
	V_{LO}	111	K3 (pos. 2)	3	E81	"	V_{LO} = E81		0.4	"
5 T_C = +125°C	f_{OSC2}	112	K5, K7, K9	13	f6	kHz	f_{OSC2} = [f6] ÷2	36	44	kHz
		113 to 128								
6 T_C = -55°C		129 to 145								
7 T_C = +25°C	N_O	146		18	E106	µV rms	N_O = E106		200	µV rms
	G_{BW}	147					$G_{BW} = f_{e1}$ when $e_o = e_i$ see figure 7	2		MHz

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers																	
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
			(+) ERROR	(-) ERROR	COMP	C_{SS}	RESET	-C.S.	+C.S.	SHUT-DOWN	R_T	C_T	R_D	SYNC	O/P A	V_C	GND	O/P B	V_{IN}	V_{REF}
9 T_C = +25°C	t_{PW}	148	GND	GND				GND	GND		4.12 kΩ	0.01 µF	0 Ω	2.7 kΩ to V_{REF}			GND		15 V	
	$t_{SD(A)}$	149	150 mV	"				"	"			11/				15 V	"		"	
	$t_{SD(B)}$	150	150 mV	"				"	"			11/				"	"		"	
	$t_{\frac{ON(A)\ MAX}{fOSC}}$	151	GND	"	3.6 V			"	"		4.12 kΩ	0.01 µF	0 Ω			"	"		"	
	$t_{\frac{ON(A)\ MIN}{fOSC}}$	152	"	"	0.4 V			"	"		"	"	"			"	"		"	
	$t_{\frac{ON(B)\ MAX}{fOSC}}$	153	"	"	3.6 V			"	"		"	"	"			"	"		"	
	$t_{\frac{ON(B)\ MIN}{fOSC}}$	154	"	"	0.4 V			"	"		"	"	"			"	"		"	
	$t_{R(tr)A}$	155	"	"	"			"	"		"	"	"			"	"		"	
	$t_{R(tf)A}$	156	"	"	"			"	"		"	"	"			"	"		"	
	$t_{R(tr)B}$	157	"	"	"			"	"		"	"	"			"	"		"	
	$t_{R(tf)B}$	158	"	"	"			"	"		"	"	"			"	"		"	
10 T_C = +125°C	t_{PW}	159	"	"				"	"		"	"	"	2.7 kΩ to V_{REF}			"		"	
	$t_{SD(A)}$	160	150 mV	GND				"	"			11/				15 V	"		"	
	$t_{SD(B)}$	161	150 mV	GND				"	"			11/				15 V	"		15 V	
		162 to 169	All test parameters, test conditions, equations, and test limits remaining are identical with tests 151 through 158 specified in table III, subgroup 9, T_C = +25°C.																	
11 T_C = -55°C		170 to 180	All test parameters, test conditions, equations, and test limits are identical with those specified in table III, subgroup 9, T_C = +25°C.																	

See footnotes at end of table.

TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Energized relays	Measured pin			Equations and notes	Limits		Unit
				No.	Value	Unit		Min	Max	
9 T_C = +25°C	t_{PW}	148	K5, K7, K9	12	t1	µs	t_{PW} = t1 16/		2	µs
	$t_{SD(A)}$	149		8 to 13	t2	"	$t_{SD(A)}$ = t2 16/		0.5	"
	$t_{SD(B)}$	150		8 to 16	t3	"	$t_{SD(B)}$ = t3 16/		0.5	"
	$\frac{t_{ON(A)\ MAX}}{t_{OSC}}$	151	K3 (pos. 2), K5, K7, K8, K9, K11, K12	13	t4, t_{per}	"	$\frac{t_{ON(A)\ MAX}}{t_{OSC}} = \frac{t4}{t_{per}} \times 100$ 16/	45	50	%
	$\frac{t_{ON(A)\ MIN}}{t_{OSC}}$	152	"	13	t5, t_{per}	"	$\frac{t_{ON(A)\ MIN}}{t_{OSC}} = \frac{t5}{t_{per}} \times 100$ 16/		0.001	"
	$\frac{t_{ON(B)\ MAX}}{t_{OSC}}$	153	"	16	t6, t_{per}	"	$\frac{t_{ON(B)\ MAX}}{t_{OSC}} = \frac{t6}{t_{per}} \times 100$ 16/	45	50	"
	$\frac{t_{ON(B)\ MIN}}{t_{OSC}}$	154	"	16	t7, t_{per}	"	$\frac{t_{ON(B)\ MIN}}{t_{OSC}} = \frac{t7}{t_{per}} \times 100$ 16/		0.001	"
	$t_{R(tr)A}$	155	"	13	t8	"	$t_{R(tr)A}$ = t8		0.3	µs
	$t_{R(tf)A}$	156	"	13	t9	"	$t_{R(tf)A}$ = t9		0.2	"
	$t_{R(tr)B}$	157	"	16	t10	"	$t_{R(tr)B}$ = t10		0.3	"
	$t_{R(tf)B}$	158	"	16	t11	"	$t_{R(tf)B}$ = t11		0.2	"
10 T_C = +125°C	t_{PW} B	159	K5, K7, K9	12	t12	"	t_{PW} = t12 16/		3	"
	$t_{SD(A)}$	160		8 to 16	t13	"	$t_{SD(A)}$ = t13 16/		0.7	"
	$t_{SD(B)}$	161		8 to 13	t14	"	$t_{SD(B)}$ = t14 16/		0.7	"
		162 to 169								
11 T_C = -55°C		170 to 180								

See footnotes at end of table.

NOTES:

1/ This test applies to device type 02 only.

2/ 5 V, 50 µs pulse, toggles the "on state" between transistors A and B.

3/ Vary V_{CM} from 1.5 V to 5.2 V and measure change on pin 9, V_N = 2.5 V.

Equation for C_{MR} test is: $C_{MR} = 20 \text{ Log} \left| \frac{\Delta V_{CM}\ (101)}{(\Delta EX - \Delta V_{CM})} \right|$

Test 18, $\Delta EX = \Delta E16$

Test 40, $\Delta EX = \Delta E35$

Test 61, $\Delta EX = \Delta E53$

4/ Load resistance on outputs A and B $\geq$ 1 kΩ.

5/ 0.8 V to 2.8 V transition on V_{SYNC} must cause a transition in V_{OSC}. 2.8 V to 0.8 V transition on V_{SYNC} must not cause a transition in V_{OSC}.

6/ Apply 0.5 V to the shutdown pin, output should be on. (Device type 02 either output > 2.5 V) (Device type 04 either output < 2.5 V).

7/ Apply 1.6 V to the shutdown pin, output should be off. (Device type 02 both outputs < 2.5 V) (Device type 04 both outputs > 2.5 V).

8/ Equation for P_{SRR} is as follows with V_N = 5.1 V:

$$P_{SRR} = 20 \text{ Log} \left| \frac{\Delta V_{IN}\ (101)}{(EX - EY)} \right|$$

Test 81, (EX - EY) = (E61 - E62)

Test 96, (EX - EY) = (E67 - E68)

Test 112, (EX - EY) = (E73 - E74)

9/ Increase V_{IN} = from 5 V to the point where outputs are on: (Device type 02 onstate = either output > 2.5 V), (device type 04 onstate = either output < 2.5 V).

10/ Device type 02, $t_{SD} = t_{output(low)} - t_{pin\ 10(high)}$.

Device type 04, $t_{SD} = t_{output(high)} - t_{pin\ 10(high)}$.

11/ Apply a 3.6 V, 50 µs pulse to C_T to switch output transistor (which is on).

12/ V_N = 2.5 V. Equation for C_{MR} is as follows:

$$C_{MR} = 20 \text{ Log } \left[\frac{5.2 \text{ V } (201)}{|EX - EY - 5.2|} \right]$$

Test number 19: EX = E17, EY = E16
Test number 50: EX = E40, EY = E39
Test number 81: EX = E63, EY = E62

13/ V_N = 0.5 V to 3.5 V. Equation for A_{VS} is as follows:

$$A_{VS} = 20 \text{ Log } \left(\frac{\Delta V_N (201)}{|\Delta EX|} \right)$$

Test number 101: ΔEX = E73
Test number 118: ΔEX = E84
Test number 135: ΔEX = E95

14/ V_N = 2.5 V. Equation for P_{SRR} is as follows:

$$P_{SRR} = 20 \text{ Log } \left[\frac{\Delta V_{IN} (201)}{|(EX - EY)|} \right]$$

Test number 102: (EX - EY) = (E75-E74)
Test number 119: (EX - EY) = (E86-E85)
Test number 136: (EX - EY) = (E97-E96)

15/ Increase voltage on +C.S. until V shutdown ≤ 0.4 V. Record V +C.S.

16/ See figure 6.

... (steady state power and reverse bias) and operating life test.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroup 8 shall be omitted.

b. Electrical tests requirements shall be as specified in table II herein.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 3.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table IV herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. Steady state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

(1) Test condition C, using the circuit shown on figure 3; or test condition F, using the circuit shown on figure 4.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Special subgroups shall be added to group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits of subgroups 10 and 11 as specified in table III herein.

TABLE IV. Group C end-point electrical parameters T_C = +25°C.

Device type	Test	Limits		Unit	Delta	Unit
		Min	Max			
All	V_{IO}	---	5	mV	±0.7	mV
All	I_{IB}	0.1	10.0	µA	±1	µA
01	V_{REF}	4.8	5.2	V	±20	mV
02,04	V_{REF}	5.0	5.2	V	±20	mV
03	V_{REF}	4.9	5.1	V	±20	mV

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the external zero reference level of the supply voltage. Current values given are for conventional current and are positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition at room temperature prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Part or Identifying Number (PIN) (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to acquiring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	1524
02	1525A
03	1526
04	1527A

6.6 Part or Identifying Number (PIN). The PIN shall be in accordance with MIL-M-38510.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1150)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/126B	MICROCIRCUITS, LINEAR, REGULATING, PULSE-WIDTH MODULATOR, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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